

4

Newton's Laws of Motion & Friction

Dekh bhai, is chapter mein Mains or advance weightage dhoondhne ki galti mat karna. Mains mein har saal isse questions puche jaate hain, but advance mein bahut hi kam questions puche gaye hain, but puri mechanics ka ye fundamental chapter hai jo puri physics mein use hoga, so is chapter ko halke mein na le



INERTIA

- ★ It is the tendency to resist the change.
- ★ It is the property due to which a body wants to be in state of rest or in uniform motion in a straight line.
- ★ Mass is the measurement of inertia.

Momentum

It is the net motion contain in the system.

$$\vec{P} = m\vec{v}$$

Dekh bhai, baad mein hum yeh dekhenge ki kisi body ya system ka momentum nikaalne ka tarika hota hai

$$\left\{ \begin{array}{l} \vec{P}_{\text{body}} = M_{\text{body}} \vec{v}_{\text{com of body}} \\ \vec{P}_{\text{system}} = M_{\text{syst.}} \vec{v}_{\text{com of syst.}} \end{array} \right\} \text{Com mein}$$



Force → Push or Pull

Types of Force

1. Gravitational force → $mg, \frac{Gm_1 m_2}{r^2}$

2. Electromagnetic force

Tension force, Normal force, Friction, Electrostatic force, magnetic force etc.

3. Strong nuclear force

4. Weak nuclear force

yah ham 12th class mein padhenge

Newton 1st Law (Law of inertia)

- ★ A body continue to be in state of rest or in state of uniform velocity (st. line) until or unless net external force acts on it.

Newton Second Law

$$\left(\vec{F}_{\text{net}} \right)_{\text{ext}} = \frac{d\vec{p}}{dt} \quad \vec{P} = m\vec{v}$$

If $m \rightarrow \text{Const}$

$$\left(\vec{F}_{\text{net}} \right)_{\text{ext}} = m\vec{a}$$

$$\left(\vec{F}_{\text{net}} \right)_{\text{ext}} = \frac{d(m\vec{v})}{dt} = m \frac{d\vec{v}}{dt} + \vec{v} \frac{dm}{dt}$$

If mass of system is constant then

$$\left(\vec{F}_{\text{net}} \right)_{\text{ext}} = \frac{m d\vec{v}}{dt} = m\vec{a}$$

Abe, iska matlab hai har jagah $F = ma$ mat laga dena. Jaise rocket propulsion mein mass variable hota hai to wahan $F = ma$ nahi laga sakte



Newton Third law

- ★ For every action there is equal and opposite reaction at the same time, simultaneously of the same nature.

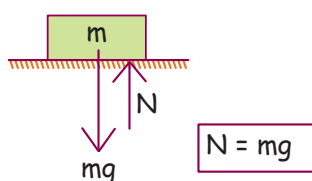
★ Action and reaction simultaneously lgte h isliye we can't say ki kon action hai aur kon reaction.

★ Two different body event.

SKC

According to this law, agar A ne B par force lagaya F, to B bhi A par wapas force lagayega F, of same nature at same time."

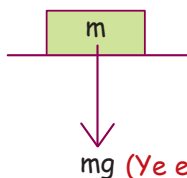
"(Is line ko yaad karlo, bahut kaam aayegi



yha normal aur mg action reaction pair nahi hai

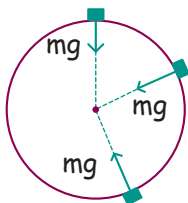
chlo pahle mg, normal, tension force padh lete hai

GRAVITATIONAL FORCE



mg (Ye earth ne mass par lgaya)

Waise toh mg ki direction towards the centre of earth hai jise ham gravitation mein detail mein padhenge.



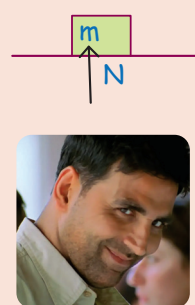
Block chahe earth par ho ya hawa mein ho mg force to lgaga hi

NORMAL FORCE

★ It act towards the body perpendicular to the surface.

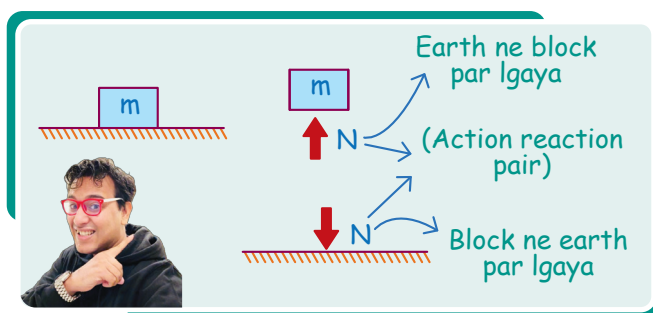
★ Pushing nature.

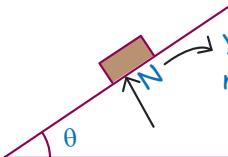
Agar koi body kisi ke saath chhu-an-chhu-ai kar rahi hai, matlab kisi surface ke saath contact/touch kar rahi hai, toh uss body par body ki taraf normal force lagega. Ab ye mat kehna ki mg kyu nahi lagaya, because abhi hum only normal force ki baat kar rahe hai.

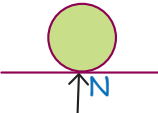
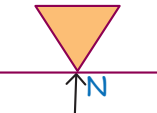


(a)  Yha earth ne block par N (normal force) lgaya

Block, Earth ke contact mein hai, toh Earth bhi toh block ke contact mein hai. Hence, Earth block ko touch kar rahi hai, toh Earth par bhi normal lagega jo block lagayega



(b)  Ye Block par lga incline plane ne lgaya

(c)  

Draw the FBD of Block

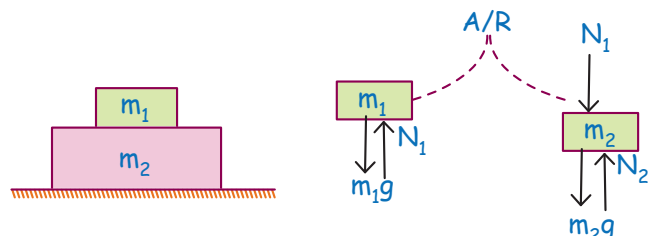
SKC

Dhyaan se sun lo, FBD (Free Body Diagram) banane ka matlab hai jis body ki hum FBD bana rahe hain, us par lagne wale saare external forces dikhane hain. Matlab, woh body doosre logon par kitna force laga rahi hai, humein isse koi matlab nahi.



Q. Draw FBD of all the block in following cases.

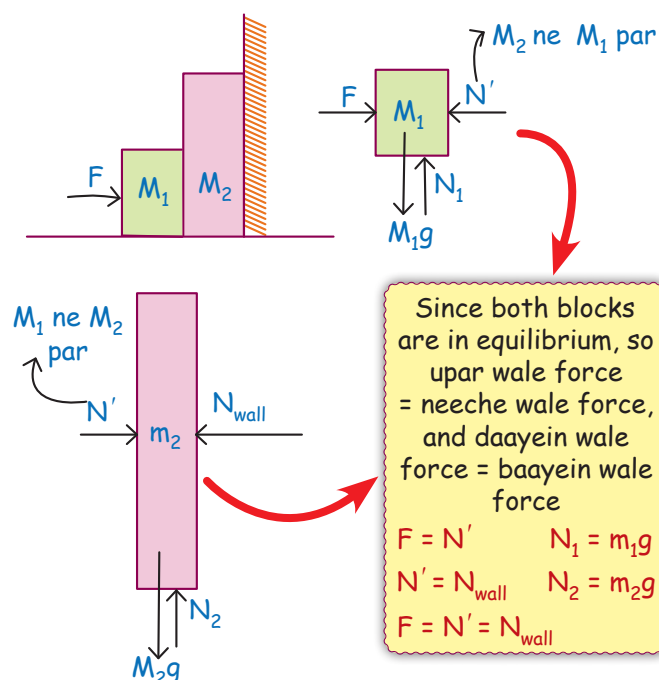
(a)



Yahan dono block equilibrium mein hain, toh har block par we can say, net upar wala force = net neeche wala force

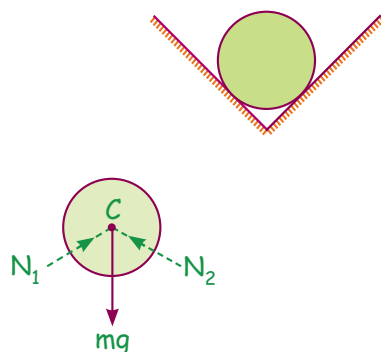
$$N_1 = m_1g \quad N_2 = N_1 + m_2g \Rightarrow N_2 = m_1g + m_2g$$

(b)



Q. A cylinder of weight W is resting on a V-groove as shown in figure. Draw its free body diagram.

Sol.

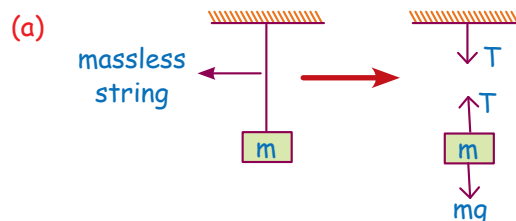


TENSION FORCE

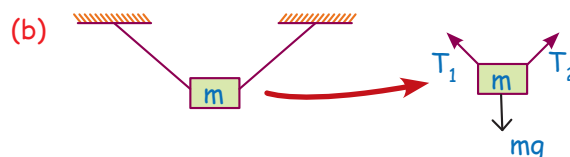
★ It act away from the body towards string.

★ It is a pulling force.

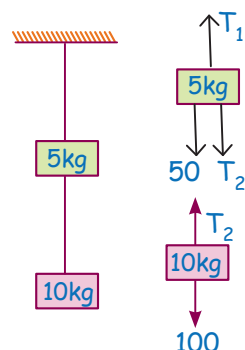
Q. Draw FBD of mass in following case.



Here if string is massless then tension at every point is massless



Q. Find tension in each string if all string are mass less.



$$T_1 = T_2 + 50 \dots (i)$$

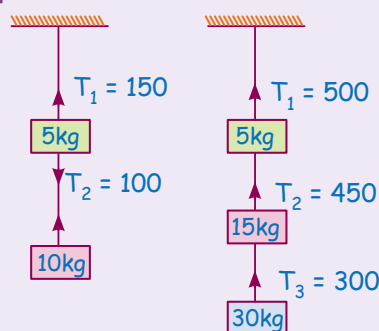
Both block are in equilibrium so upar wala force = niche wala force

$$T_2 = 100 \dots (ii)$$

Solve (i) and (ii)

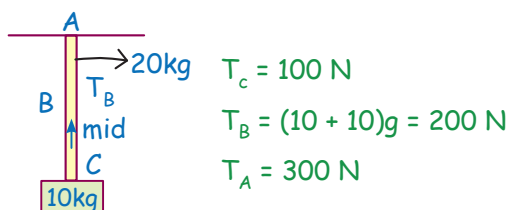
$$T_1 = 150$$

Abe ye detail mein toh mujhe majboori mein likhna pad raha hai, bhai tu toh isse direct kar. Upar wali rasssi ko pakad, dekh neeche kitna mass latka hai (15 kg so, $T_1 = 150$) similarly neeche wali rasssi ko pakad kar tension nikalo

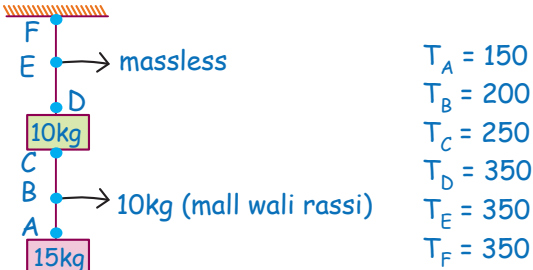


Q. 10 kg block is suspended through a string of mass 20 kg as shown in figure. Find T_A , T_B , T_C .

Sol. Jiss jagah tension puchi hai, dekho uske neeche total kitna mass hai.

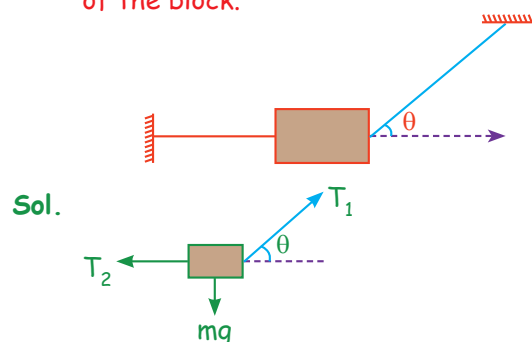


Q.



Mall wali rassi abe mtlb mass wali rassi

Q. A block of mass m is attached with two strings as shown in figure. Draw the free body diagram of the block.



EQUILIBRIUM wale sawal

SKC

Agar body equil (equilibrium) mein hai, toh sabse pehle uski FBD banao, saare forces ko X-Y axis mein tod do, aur phir $F_{net} = 0$ kar do. Matlab upar wale force = neeche wale force, left wale force = right wale force

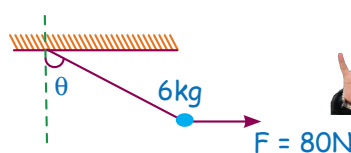
If body is in equil

$\Rightarrow \vec{F}_{net} = 0$ (Translational equil.)

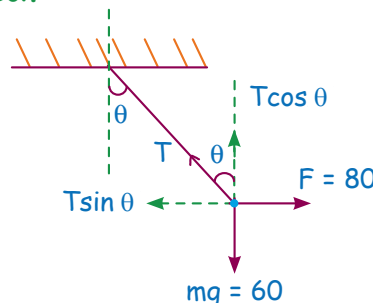
$\Sigma \vec{F}_x = 0, \Sigma \vec{F}_y = 0, \Sigma \vec{F}_z = 0$



Q. Body is in equilibrium find $\theta = ?$



Sol.



Bhai hasna mat, yeh question pichle 5 saal mein 4 baar Mains ne pucha hai

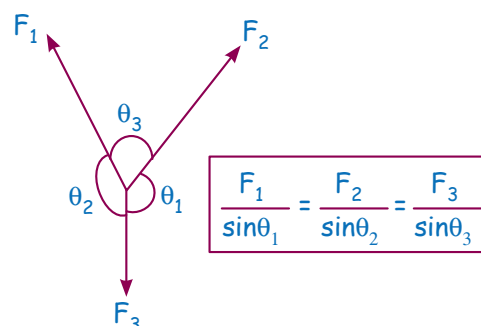
$$\begin{aligned} T \sin \theta &= 80 \\ T \cos \theta &= 60 \\ \tan \theta &= \frac{80}{60} = \frac{4}{3} \end{aligned}$$

$$\begin{aligned} \theta &= 53 \\ T \sin 53^\circ &= 80 \\ T &= 100 \end{aligned}$$

or after solving we got $T = \sqrt{F^2 + (mg)^2}$

M-2 Lami theorem

If $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0$, in following case.



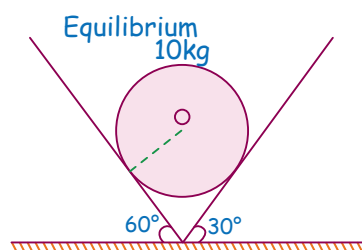
Let's solve last ques from lami theorem

$$\begin{aligned} \frac{80}{\sin(180 - \theta)} &= \frac{T}{\sin 90} = \frac{60}{\sin(90 + \theta)} \\ \frac{80}{\sin \theta} &= \frac{60}{\cos \theta} \\ \tan \theta &= \frac{80}{60} \Rightarrow \theta = 53 \end{aligned}$$

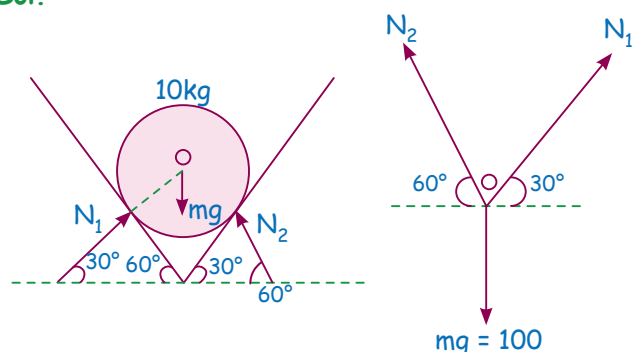
Bhai mujhe toh ye Lami's theorem ghatiya method lagta hai because na toh isse feel aati hai aur time utna hi lagega. Maine yahan bata diya hai kahin tu baad mein bole bataya nahi. So, I will suggest ki component leke, force equate karke solve karo



Q. Sphere is in equil find force applied by inclined plane in following figure.



Sol.

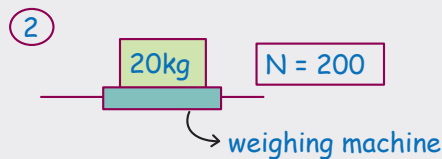
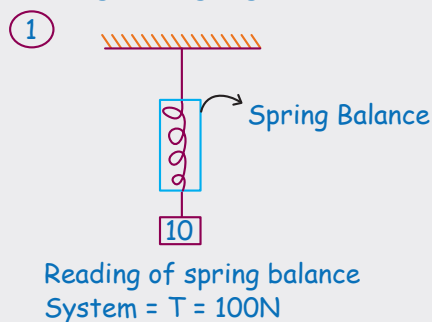


$$N_1 \cos 30 = N_2 \cos 60$$

$$N_2 \sin 60 + N_1 \sin 30 = mg = 100$$

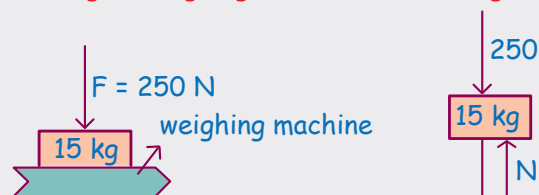
Now you can solve and get N_1 and N_2 aur man kre toh lami theorem se answer verify kr lena

Dekh bhai, question mein string mein tension puchne ka ek tarika hota hai 'reading of spring balance'. Similarly, normal puchne ka ek tarika hota hai 'reading of weighing machine'



Reading of weighing machine = $N = 200$ newton

Q. Reading of weighing machine in following.

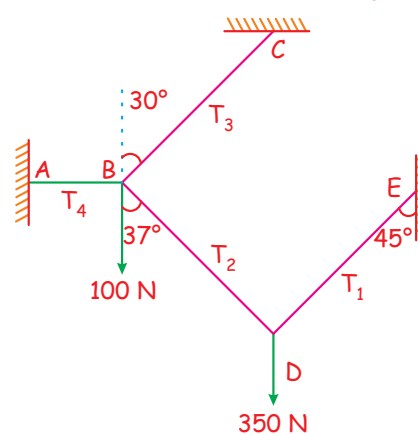


Reading of weighing machine will be

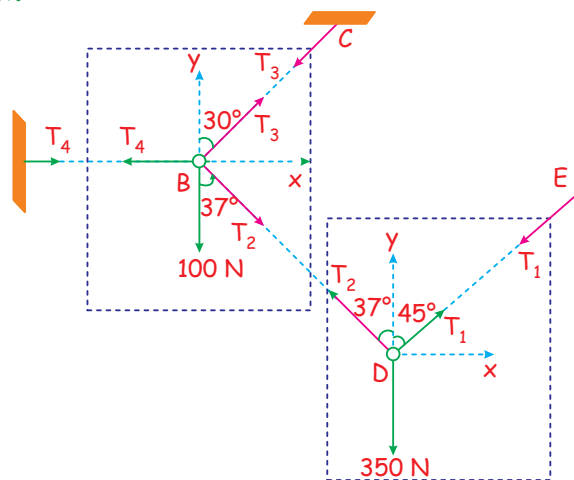
$$N = 250 + 150 = 400\text{N}$$

① 25 kg ② 20 kg ③ 40 kg ④ 50 kg

Q. Two particles of masses 10 kg and 35 kg are connected with four strings at points B and D as shown in figure. Find T_1, T_2, T_3



Sol.



Analysing the equilibrium of point D:

$$\sum F_x = 0 \text{ or } T_1 \sin 45^\circ - T_2 \sin 37^\circ = 0 \quad \dots (i)$$

$$\text{and } \sum F_y = 0 \text{ or } T_1 \cos 45^\circ + T_2 \cos 37^\circ = 350 \quad \dots (ii)$$

Solve both eqn. and get

$$\Rightarrow T_1 = 150\sqrt{2}\text{N and } T_2 = 250\text{N}$$

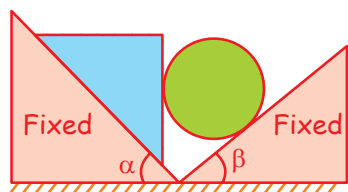
Similarly for equil of point B
 $T_3 \cos 30^\circ = 100 + T_2 \cos 37^\circ$
 $T_3 \sin 30^\circ + T_2 \sin 37^\circ = T_4$
 Put value of $T_2 = 250$,
 & solve both eqn get
 the ans.



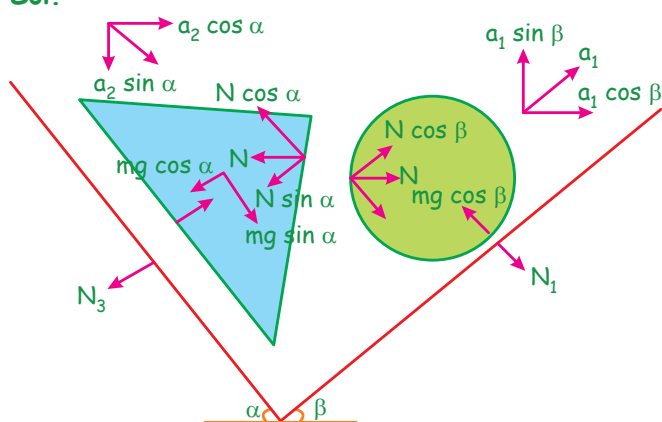
Abe, kyun ro raha hai? Mujhe pata hai calculation mushkil hai, lekin advance ke point of view se aadat daal le. Ye le, do sawal aur le



Q. A cylinder and a wedge of same masses with a vertical face, touching each other, move along two smooth inclined planes forming the same angle α and β respectively with the horizontal. Determine the force of normal N exerted by the wedge on the cylinder, neglecting the friction between them.



Sol.



It is obvious that acceleration of cylinder is parallel to the wedge 1 and acceleration of triangular block is parallel to the wedge 2.

$$a_2 \cos \alpha = a_1 \cos \beta \quad \dots(i)$$

[constrained relation between the contact surface of block and cylinder]

$$N \cos \beta - m_1 g \sin \beta = m_1 a_1 \quad \dots(ii)$$

[Newton's II law for cylinder along the direction parallel to the wedge 1]

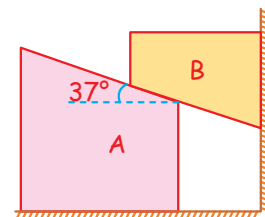
$$m_2 g \sin \alpha - N \cos \alpha = m_2 a_2 \quad \dots(iii)$$

[Newton's II law for block along the direction parallel to the wedge 2]

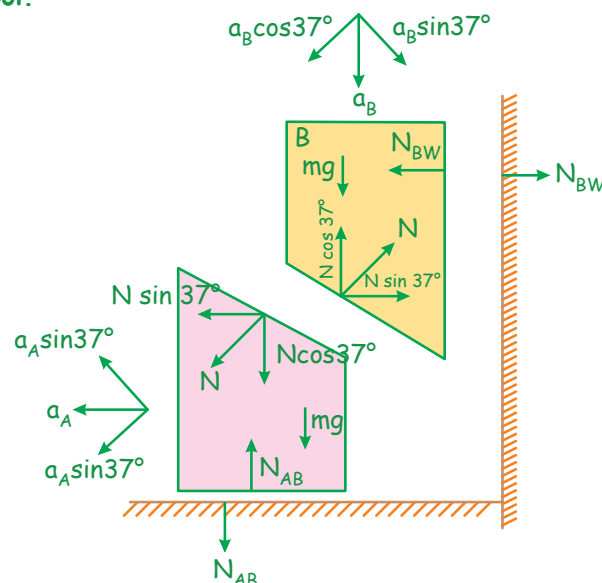
By solving equation (i), (ii) and (iii) we get

$$N = mg \left(\frac{\sin \alpha \cos \alpha + \sin \beta \cos \beta}{\cos^2 \alpha + \cos^2 \beta} \right)$$

Q. The masses of blocks A and B are same and equal to m . Friction is absent everywhere. Find the magnitude of normal force with which block B presses on the wall and accelerations of the blocks A and B.



Sol.



$$mg - N \cos 37^\circ = m a_B$$

[Newton's II law for block B in vertical direction]

$$N \sin 37^\circ = m a_A$$

[Newton's II law for block A in horizontal direction]

$$a_B \cos 37^\circ = a_A \sin 37^\circ$$

[constrained relation for contact surface between block A and B]

By solving above three equations we get

$$a_A = \frac{12g}{25} \quad a_B = \frac{9g}{25} \quad N = \frac{4mg}{5}$$

$$N_{BW} = N \sin 37^\circ$$

[Equilibrium of block B in horizontal direction]

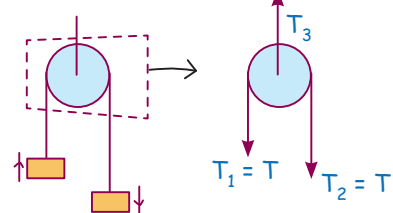
$$\Rightarrow N_{BW} = \frac{12mg}{25}$$

PULLEY SYSTEM / ATWOOD MACHINE

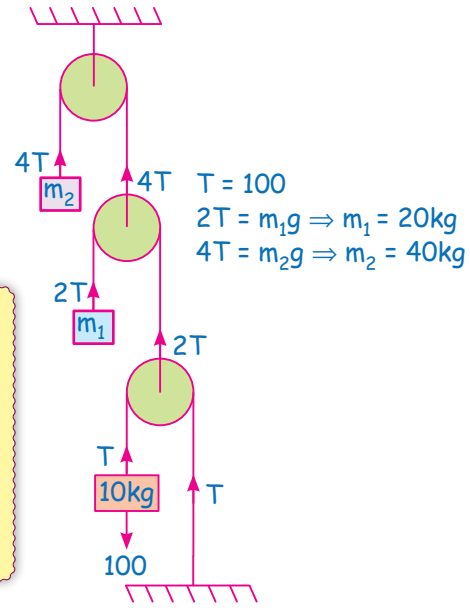
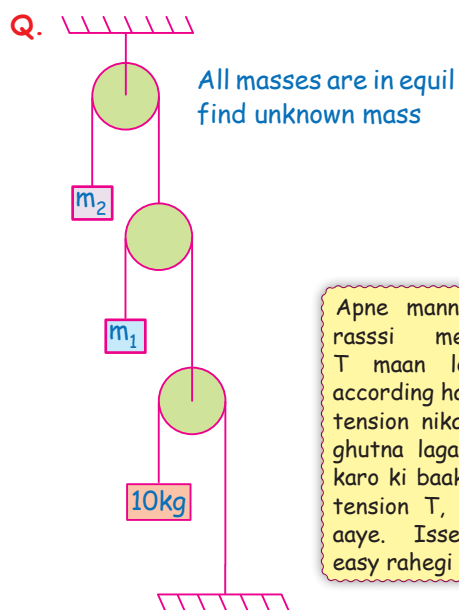
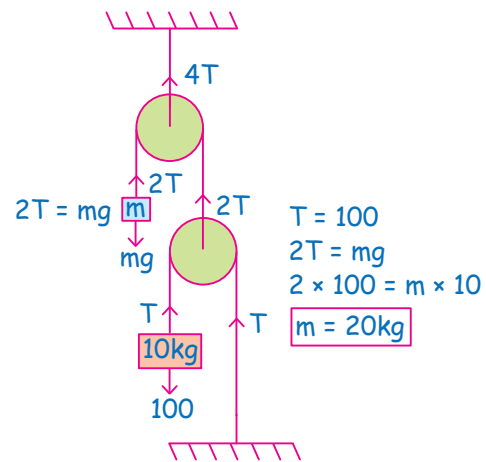
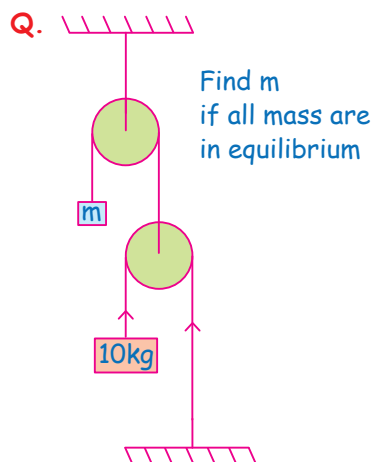
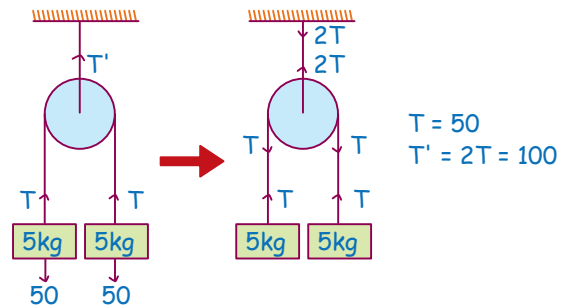
Real Pulley



Ideal Pulley → massless, no friction between string and pulley.



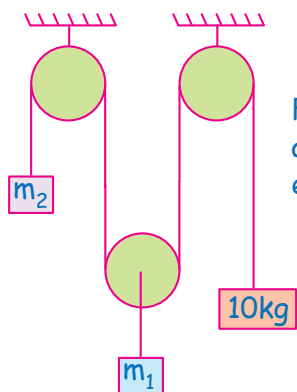
If pulley is ideal then $T_3 = T_1 + T_2 = 2T$



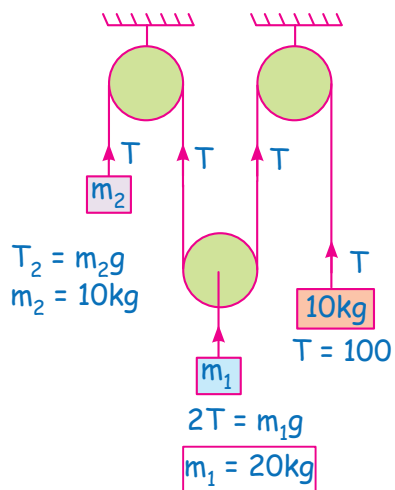
Apne mann se kisi ek rasssi mein tension T maan lo aur uske according har rasssi mein tension nikalo. Thoda sa ghutna laga kar koshish karo ki baaki rasssi mein tension T , $2T$, $4T$ type aaye. Isse calculation easy rahegi



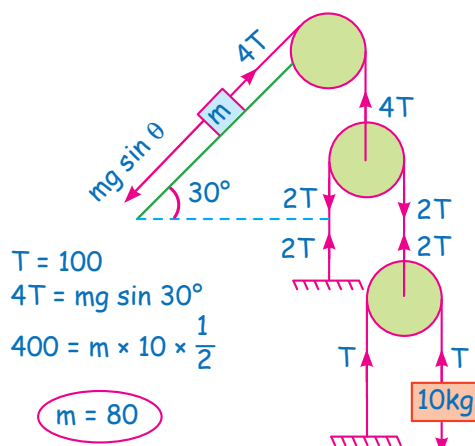
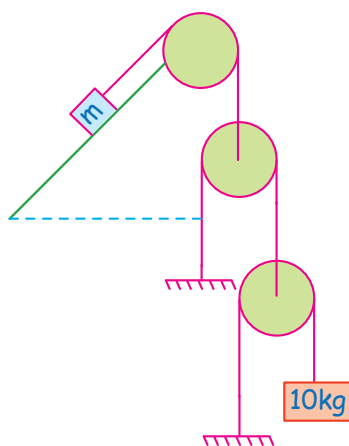
Q.



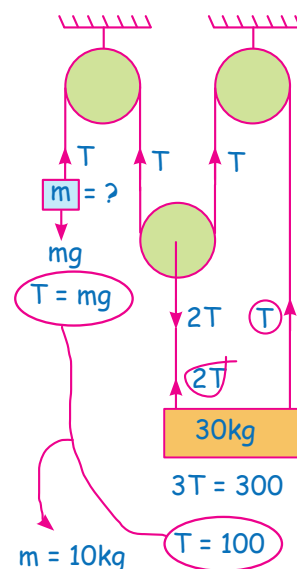
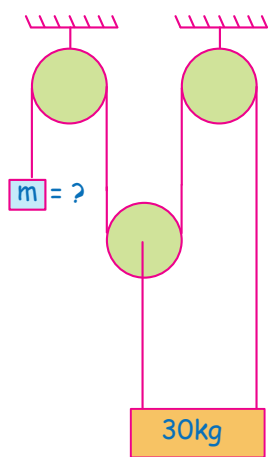
Find m_1 and m_2 if all masses are in equilibrium



Q.



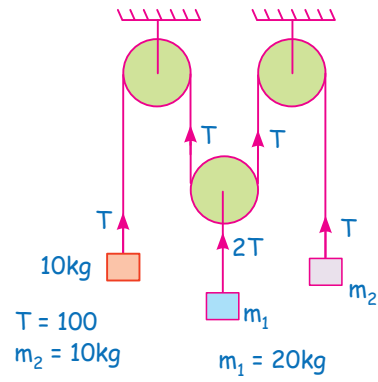
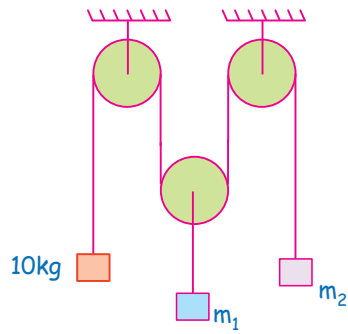
Q.



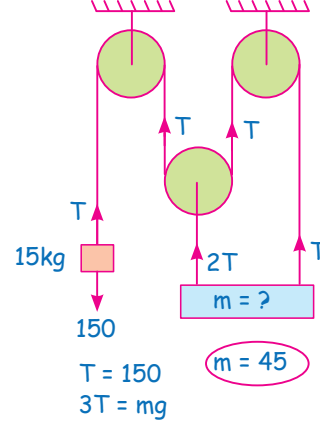
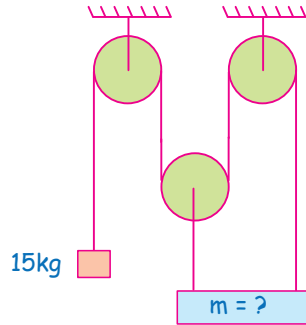
Kisi bhi rasssi mein tension T maano, aur phir carefully solve karo, final answer same hi aayega. Matlab, agar tumhari wali ne kisi doosri rasssi mein T mana hai, toh uska answer bhi sahi aayega



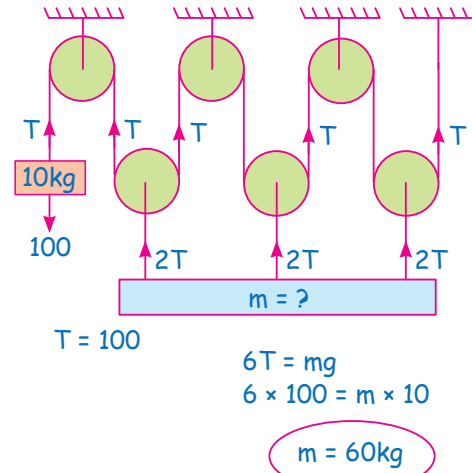
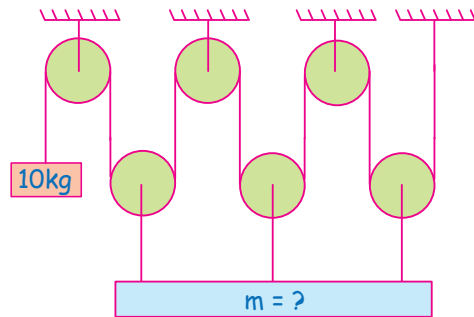
Q.



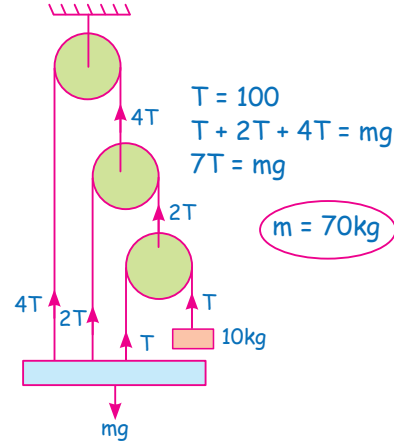
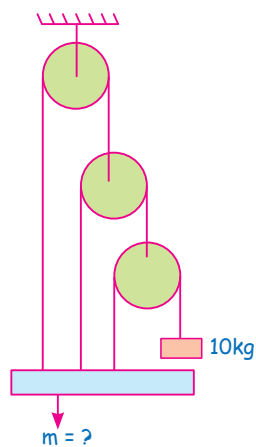
Q.



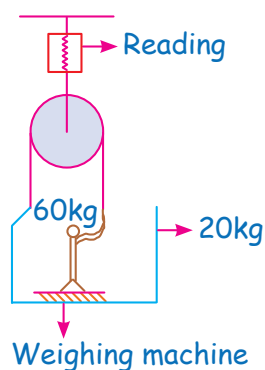
Q.



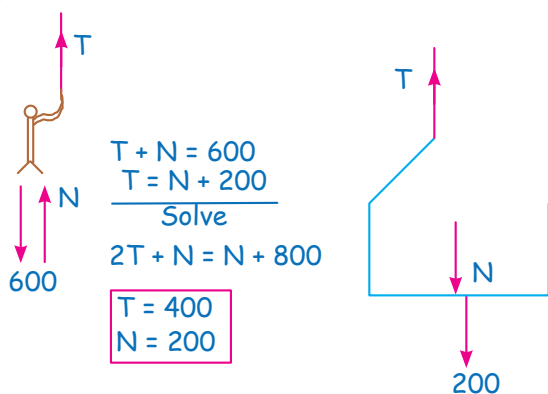
Q.



Q. Man is pulling the string by applying force F such that whole system is in air and equili. Find T , Normal contact force between man and platform



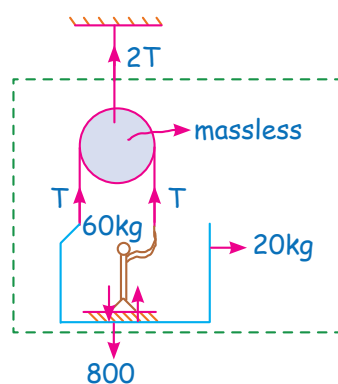
Sol.



* Reading of spring balance = $2T = 800 \text{ N}$

* Reading of weighing machine = $N = 200 \text{ N}$

Method-2: agar main pulley, man, cage ko system maan loon to system is in equilibrium.



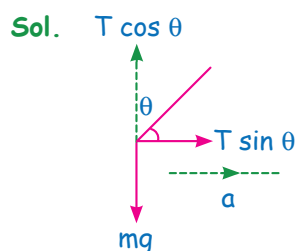
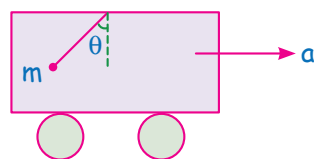
$$2T = 800$$

$$T = 400$$

Direct answer aa gya



Q. A car is moving with constant acc a , as shown in fig. if the string is making constant angle θ . Find θ .

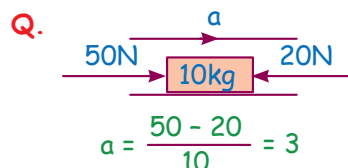
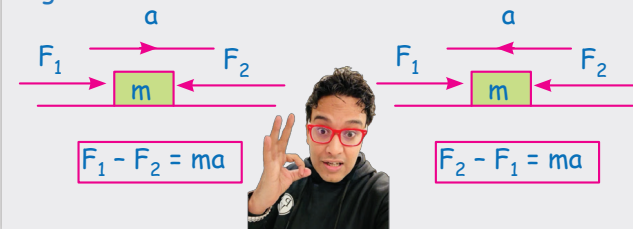


$$T \cos \theta = mg$$

$$T \sin \theta = ma$$

$$\tan \theta = \frac{a}{g}$$

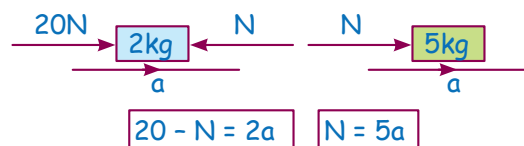
Very very simple but very very very important इसे lightly Bilkul mat lena, because ek karod baar ye use hoga



Q. Find acc of block and normal force between them.



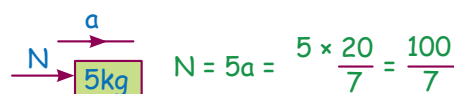
Sol.



Solve and get

$$a = \frac{20}{7}, N = \frac{100}{7}$$

Method-2:



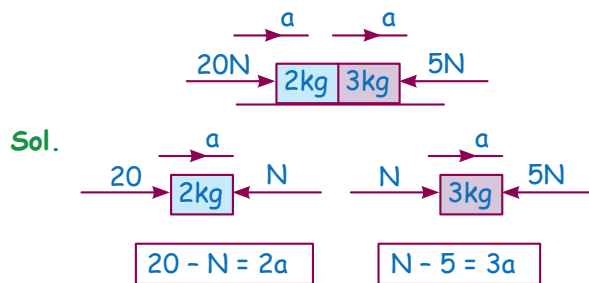
If acc of each block is same

$\vec{a} \rightarrow$ Same

$$\left(\vec{F}_{\text{net}} \right)_{\text{ext}} = M_{\text{total}} \vec{a}$$



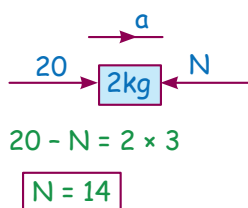
Q. Find acc and normal between blocks.



Solve and get $a = 3$, $N = 14$

Method-2:

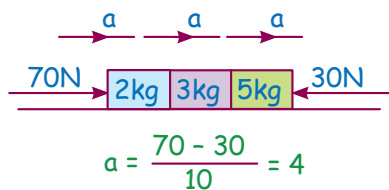
$$\begin{aligned} \vec{F}_{\text{net ext}} &= 20 - 5 = (2 + 3)a \\ 15 &= 5a \\ a &= 3 \end{aligned}$$



SKC

Bhai, jo bachche pehli baar NLM padh rahe hain, unke liye ek bahut hi important baat bata raha hoon. Jab bhi kabhi normal, tension ya spring force pucho, ye yaad rakho ki FBD banakar equation likhni hai (VVI - Very Very Important)

Q. Find acc and normal between blocks.

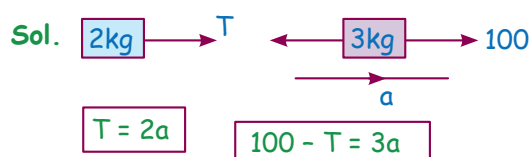


force applied by 3kg on 5kg



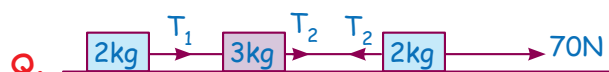
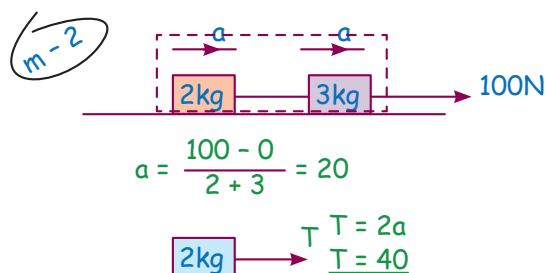
$$\begin{aligned} N - 30 &= 5a \\ N &= 30 + 20 = 50 \end{aligned}$$

Q. Find acc of each block and tension in string.



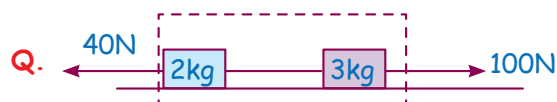
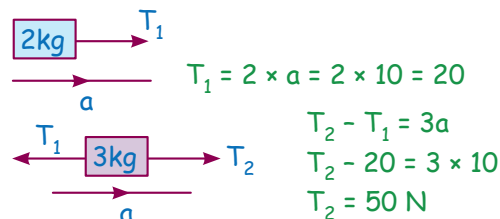
Solve and get $a = 20$ and $T = 40$

$$\begin{aligned} a &= 20 \\ T &= 40 \end{aligned}$$



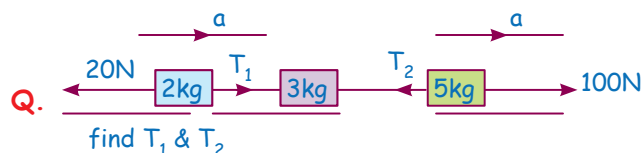
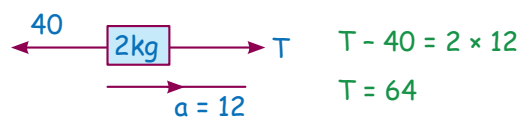
Sol.

$$a = \frac{70 - 0}{2 + 3 + 2} = 10$$



Sol.

$$a = \frac{100 - 40}{2 + 3} = 12$$



Sol.

$$\begin{aligned} a &= \frac{100 - 20}{2 + 3 + 5} = 8 \\ T_1 - 20 &= 2 \times 8 \\ T_1 &= 36 \end{aligned}$$

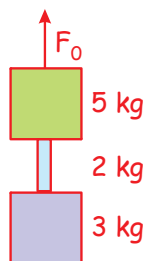
$$\begin{aligned} 100 - T_2 &= 5 \times 8 \\ T_2 &= 60 \end{aligned}$$

Q. A 5 kg block has a rope of mass 2 kg attached to its underside and a 3 kg block is suspended from the other end of the rope. The whole system is accelerated upward at an acceleration of 2 m/s^2 by an external force F_0 . ($g = 10 \text{ m/s}^2$)

(a) What is the value of F_0 ?

(b) What is the net force on the rope?

(c) What is the tension at middle point of the rope?



Sol. For calculating the value of F_0 , consider two blocks with the rope as a system.

F.B.D. of whole system

(a) $F_0 - 100 = 10 \times 2$

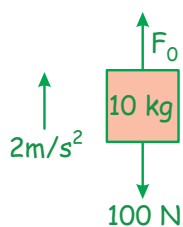
$F_0 = 120 \text{ N} \quad \dots (i)$

(b) According to Newton's second law, net force on rope

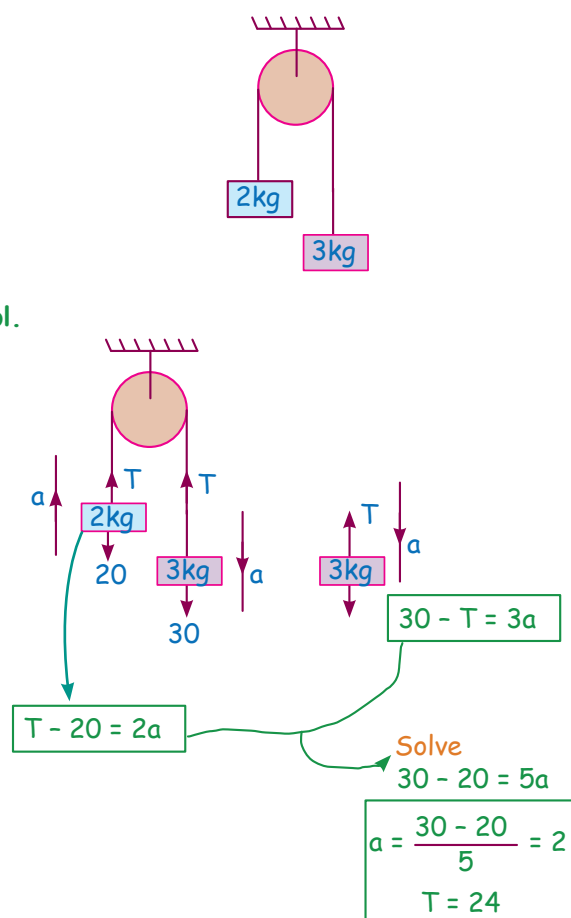
$F = ma = (2)(2) = 4 \text{ N}$

(c) For calculating tension at the middle point we draw F.B.D. of 3 kg block with half of the rope (mass 1 kg) as shown.

$T - 4g = 4 \times (2) \Rightarrow T = 48 \text{ N}$

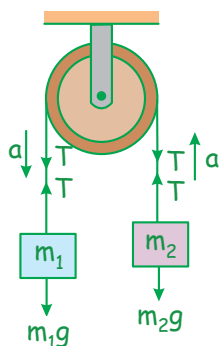


Sol.



Q. When two bodies of mass m_1 and m_2 are attached at the ends of a string passing over a pulley (neglecting the mass of pulley).

Sol.



$m_1 a = m_1 g - T$ (for m_1) $m_2 a = T - m_2 g$ (for m_2)

Solve and get (i) and (ii)

$a = \frac{(m_1 - m_2)g}{(m_1 + m_2)}, T = \left(\frac{2m_1 m_2}{m_1 + m_2} \right) g$

Q. Find acc of each mass if pulley is ideal, string is massless and there is no friction between pulley and rope.

SKC



are bhai isme ek pattern hai vo catch kr rhe ho na

$a = \frac{\text{khichne wala force} - \text{roknne wala force}}{\text{Total mass}}$

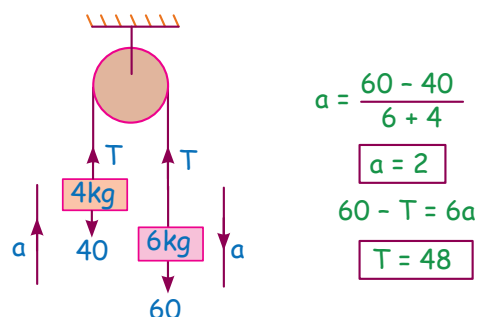
Bole to

$a = \frac{\text{Net driving force}}{\text{Total mass}}$

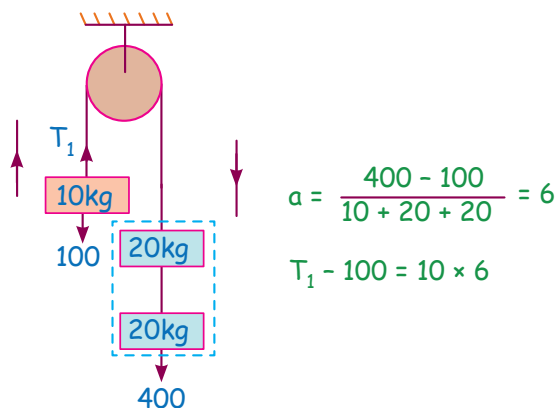
Now solve following question.



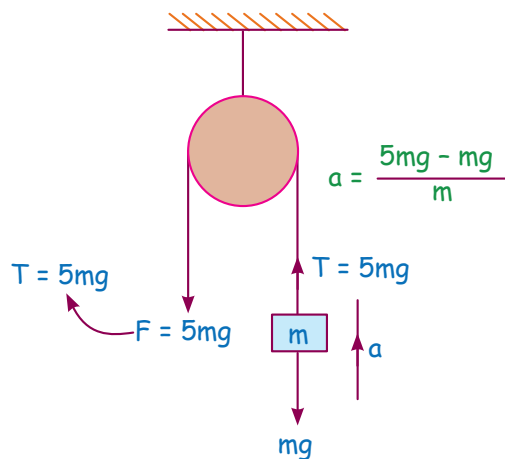
Q.



Q.



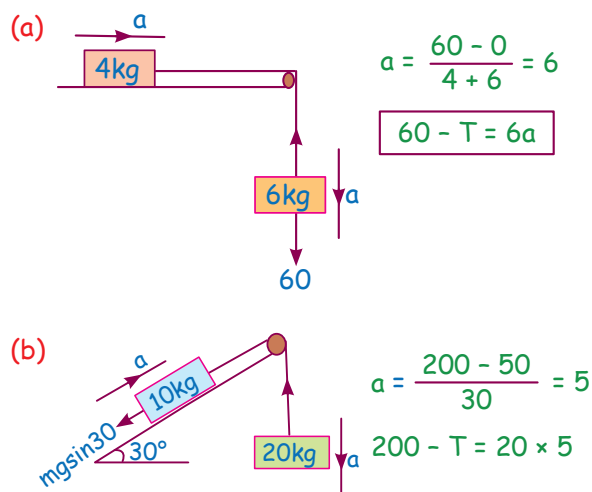
Q. Find acc of mass m.



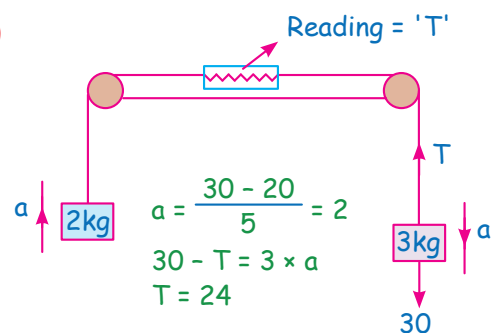
Dekh bhai, is baat ko dhyan mein rakhte hue ki ye 11th class ki book hai aur many of you are 10th pass, so, bahut patience ke saath maine is profile ke basic fundamental questions karwa diye hain. Ab neeche wale ek-ek question aapko solve karna hai



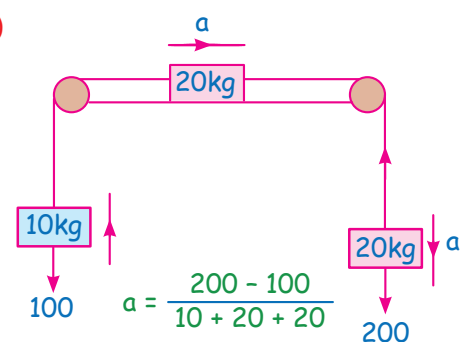
Q. Find acc and tension in following case



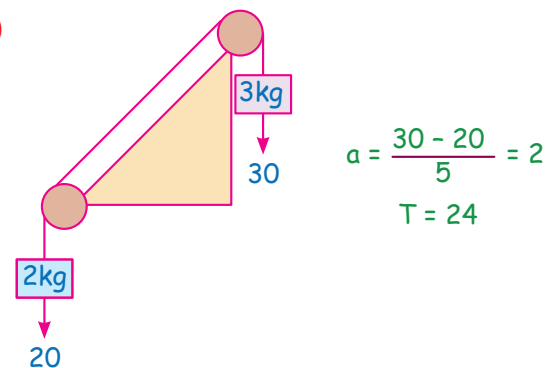
(c)



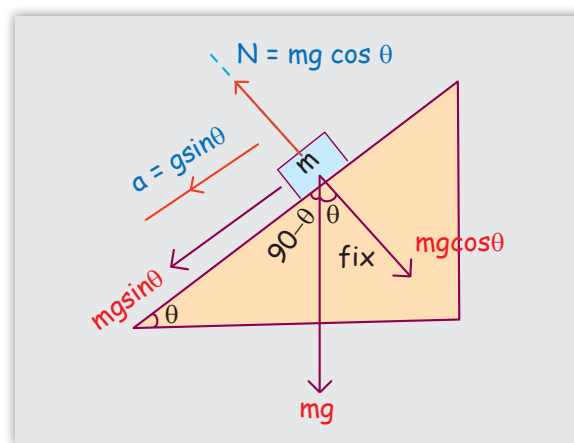
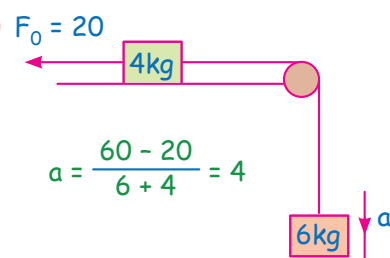
(d)



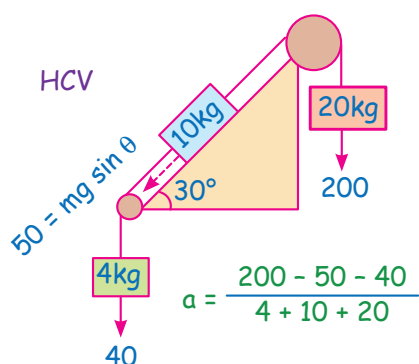
(e)



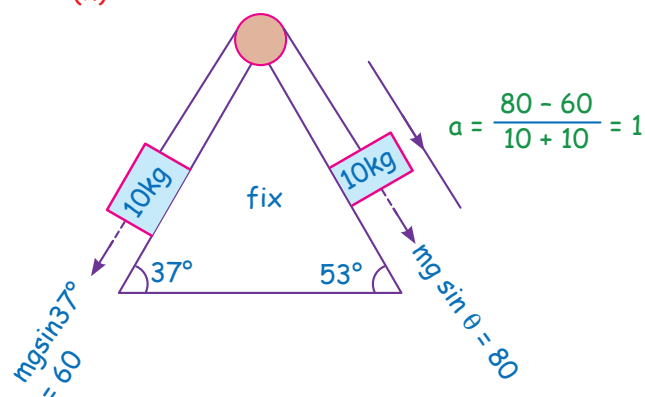
(f)



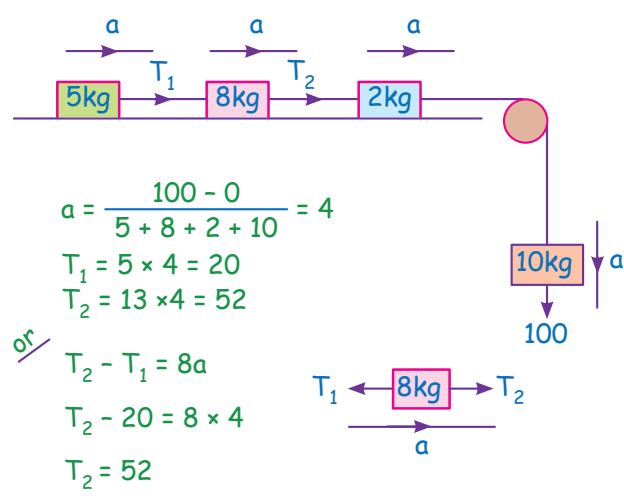
(g)



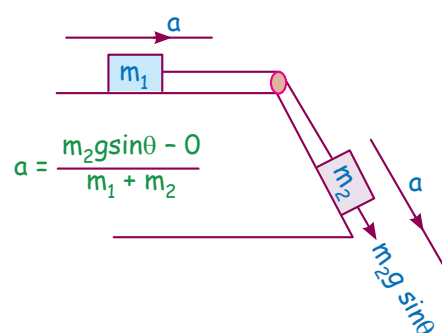
(h)



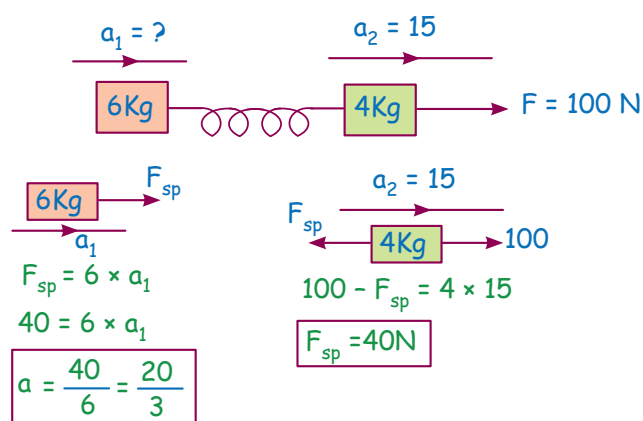
(i)



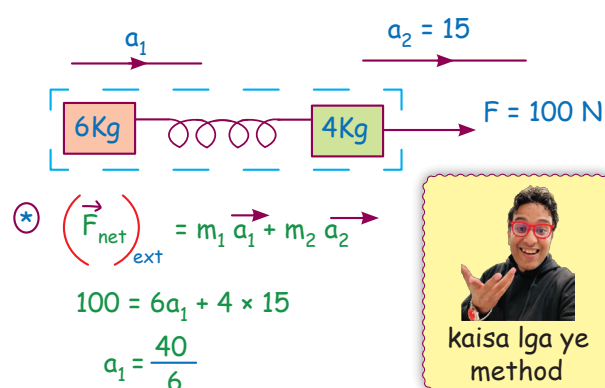
(j)



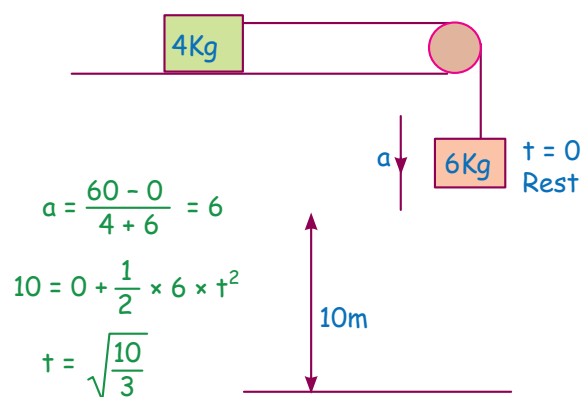
(k)



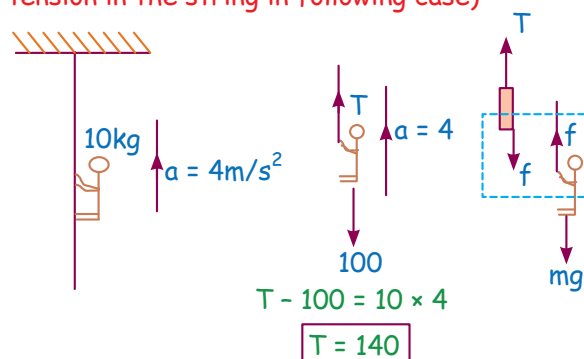
(m) Method-2:



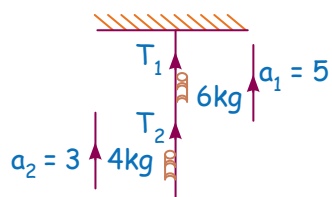
Q. Find time when 6kg block strike the ground.



Q. Find force applied by the monkey on string (i.e tension in the string in following case)



Q. Find tension in upper part of string and in middle part of string in the following figure.



Sol.

$$T_2 - 40 = 4 \times 3$$

$$T_2 = 52$$

$$T_1 - T_2 - 60 = 6 \times 5$$

$$T_1 - 52 - 60 = 30$$

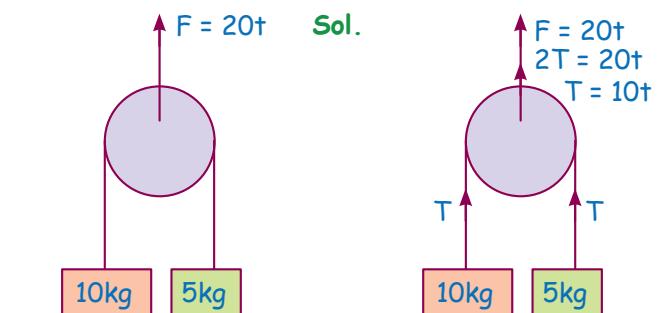
$$T_1 = 142$$

Solve and get $T_2 = 52$ and $T_1 = 142$

SKC

min value of F so that it lift off
 $mg = 100$ $F > 100$
 but हमें ans $F_{\min} = 100$ dena hai

Q. (a) When 5kg block leave the floor

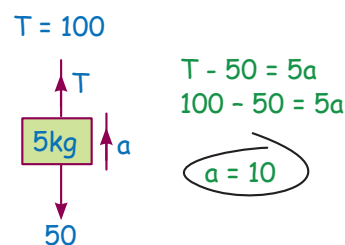


5kg kab udega jab $T = 50$
 $10t = 50 \Rightarrow t = 5$ seconds

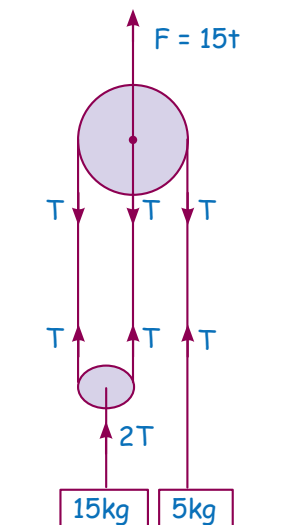
(b) When 10kg block leave the floor

$T = 100$
 $10t = 100 \Rightarrow t = 10$ sec

(c) find acceleration of 5kg block when 10kg block will leave the floor



Q. Find the time when 5kg block lift off



Sol.

$$T' = 3T$$

$$3T = 15t$$

$$5kg \text{ kab udega jab } T = 50$$

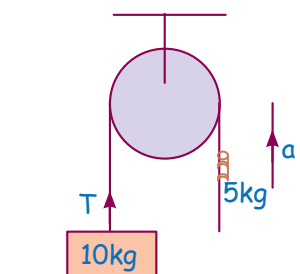
$$3 \times 50 = 15t$$

$$t = 10 \text{ sec}$$

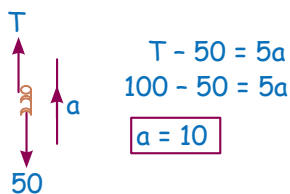
(b) when 15 kg block will lift off

$2T = 150 \Rightarrow T = 75$
 $F = 15t = 3T = 3 \times 75$
 $t = 5$ sec

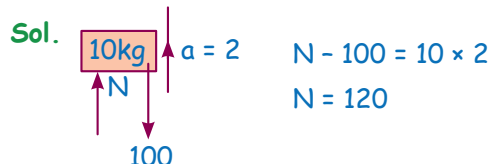
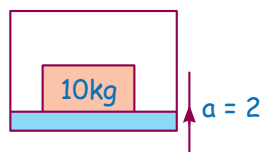
Q. What should be min ace of monkey so that block lift off the ground



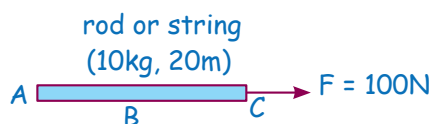
Sol. $T = 100$



Q. Find normal contact force in following figure.

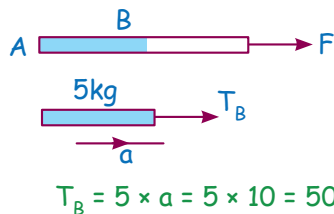


Q. Find tension at the mid point of rod



Sol. $a = \frac{100}{10} = 10$

T at point 'B'



Q. In above question find $T = f(x)$, where x is distance from end A.

Sol. (M, L)

$$a = \frac{F}{M}$$

$$T = m_x a$$

$$L \rightarrow m$$

$$T = \frac{m}{L} x \cdot \frac{F}{M}$$

$$i \rightarrow \frac{m}{L}$$

$$x \rightarrow \frac{m}{L} x$$

$$\boxed{T = \frac{Fx}{L}}$$

Q. Repeat the above problem (If $F_1 > F_2$).

Sol.

$$a = \frac{F_1 - F_2}{m}$$

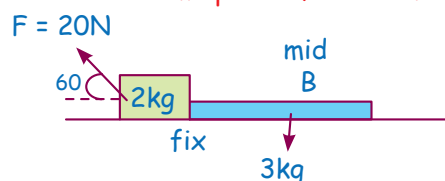
$$T = f(x)$$

$$T - F_2 = m_x a$$

$$T = F_2 + \left(\frac{m}{L} x\right) \left(\frac{F_1 - F_2}{m}\right)$$

$$T = F_2 + (F_1 - F_2) \frac{x}{L}$$

Q. Find tension at midpoint of the rod.

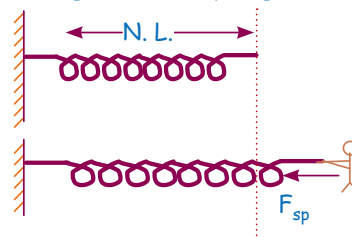


Sol. $a = \frac{F \cos 60}{2 + 3} = \frac{20 \times (0.5)}{5} = 2$

$$T_B = \frac{3}{2} \times a = \frac{3}{2} \times 2 = 3$$

IDEAL SPRING

- ★ Massless
- ★ Pitch same
- ★ N.L.
⇒ Natural length = released length
= original length
- ★ Tension throughout the spring is same.



Hook's Law

$$F_{sp} \propto -x$$

$$F_{sp} = -kx$$

elongation or compression from natural length.

Spring constant

Towards the natural length



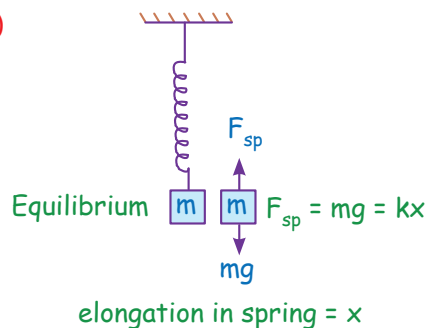
*

$$|F_{sp}| = kx$$

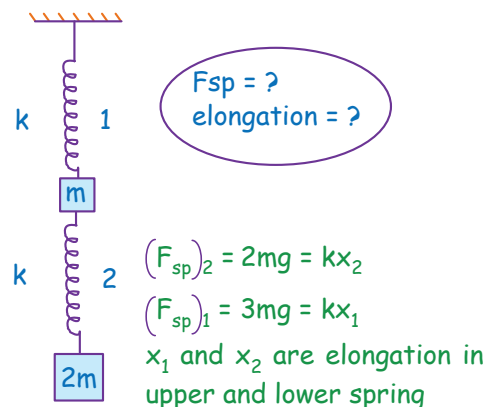
Spring ko natural length se x kheenchha ya x dabaya toh woh 'kx' force lagayega

Q. Block is in equilibrium find spring force and elongation in the spring in following cases.

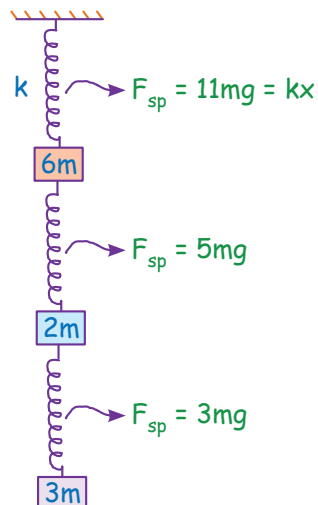
(a)



(b)



Q.



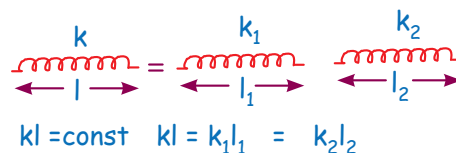
Bus spring force nikal kar use kx ke barabar kar do, toh elongation in the spring (x) pata chal jaayega



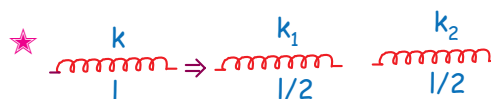
Spring Cutting

Q. A spring of length l and spring constant k is cut into two spring of length l_1 and l_2 find spring constant of both spring after cutting.

Sol.



Bus itna yaad rakho, agar material same hai toh $k \times l$ ka product same hoga



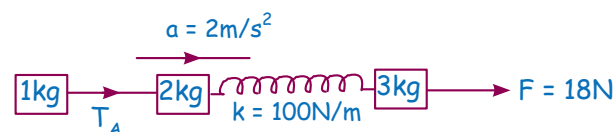
$$kl = k_1 \frac{l}{2} = k_2 \frac{l}{2}$$

$$k_1 = 2K$$

$$k_2 = 2K$$



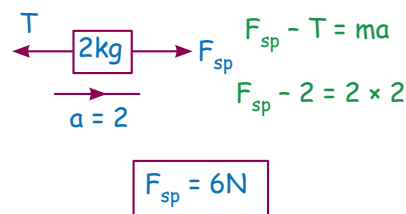
Q. (a) Find tension and acc of each block.



Sol. $a_1 = 2$

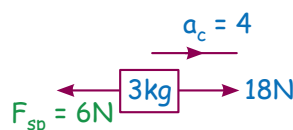
$$T_A = m_A a_A = 1 \times 2 = 2\text{N}$$

(b) Find spring force & elongation in the spring



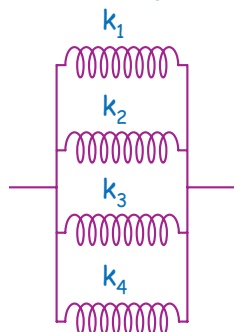
$$F_{sp} = kx \Rightarrow 6 = 100x \Rightarrow x = .06\text{m}$$

(c) Find acc of 3kg block



EQUIVALENT SPRING CONSTANT

- ★ When two springs are connected in parallel then we can replace them by single spring of spring constant k_e where $k_e = k_1 + k_2$.



Bus itna yaad rakhna ki yahan result resistance ke series/parallel ke ulte hai



For more than two spring $k = k_1 + k_2 + k_3 + \dots$

- ★ When two springs are connected in series then we can replace them by single spring of spring constant k_e

Where $1/k_e = 1/k_1 + 1/k_2$



$$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} + \dots$$

INERTIAL FRAME / NON INERTIAL FRAME

Inertial Frame

Voh frame where NLM are valid.

- ★ Rest wrt ground or moving with const velocity. wrt ground.

Non Inertial Frame (NIF)

Voh frame where NLM fails & directly not applicable.

- ★ all the accelerated frame wrt ground are NIF
- ★ Agar a hai $\Rightarrow$ Non - inertial.
- ★ In order to validate NLM eqn in a non inertial frame, we must apply a correction factor called pseudo force.

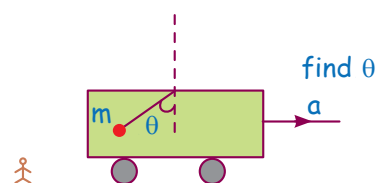
$$* \vec{F}_{\text{pseudo}} = - M_{\text{block}} \vec{a}_{\text{observer}} = - M_{\text{block}} \vec{a}_{\text{NIF}}$$

SKC

Dekh bhai, mudde ki baat ye hai ki:

1. Ground ke respect mein agar kisi frame ke paas acceleration hai, toh woh non-inertial frame hai, aur agar uss frame mein jaake hum directly Newton's Laws of Motion (NLM) ki equation likhe toh equation galat aayegi. Toh Newton ke law ki izzat bachane ke liye humne ek zabardasti ka force lagaya jo ki sachchi mein nahi lag raha hai (false force, pseudo force) taaki hum non-inertial frame mein bindass NLM equation likh sakein.
2. Pseudo force = (Mass of block) $\times$ (Acceleration of non-inertial frame), aur iski direction non-inertial frame ke acceleration ki direction ke opposite hoti hai.
3. Ye samajh lo ki ye NSP (taadne) ka license/kiraya hai. Matlab jab bhi kabhi accelerated lift, wedge, car, balloon par jaakar maje lene ho toh sabse pehle block par pseudo force laga do aur phir maja lo..... physics ka.
4. Agar aap ground par khade hokar question solve kar rahe ho toh pseudo force ki zaroorat nahi hai because in this case, pseudo force = 0.
5. Question chahe ground par khade hokar solve karo ya NIF/accelerated wedge, car mein baith kar, answer same aayega. Matlab saare questions bina pseudo force ke ground frame mein bhi solve kiye ja sakte hain."

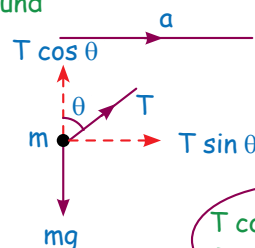
- Q. In following question find θ if block is at rest with respect to car. Analyse the situation in ground frame and also in car frame.



Sol.

Method-1: In ground frame

Wrt ground



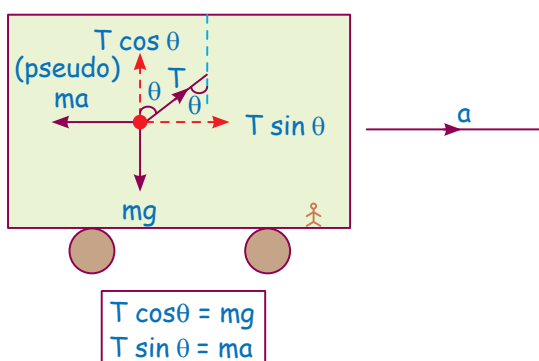
$$\begin{aligned} T \cos \theta &= mg \\ T \sin \theta &= ma \end{aligned}$$

You → sir hamesha ye confusion rehta hai, kab pseudo force lagana hai aur kab nahi?"

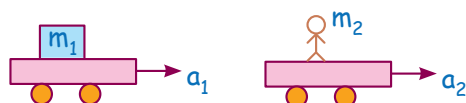


Are, jab ground par khade hokar question solve kar rahe ho toh pseudo force mat lagao, aur jab car ($a \neq 0$) ke andar baith kar question solve kar rahe ho toh pseudo force laga do

Method-2: In car frame



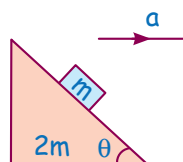
SSSQ.



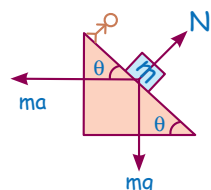
What will be the Pseudo force on ' m_1 ' wrt man?

Sol. $\vec{F}_{\text{Pseudo}} = m_1 a_2 (-\hat{i})$

Q. What should be the acceleration of wedge 'a' so that the block remains at rest wrt wedge.

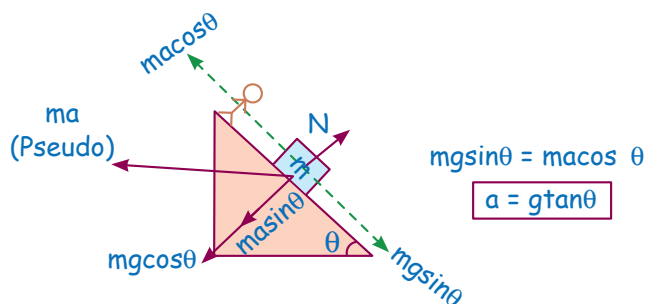


Sol.

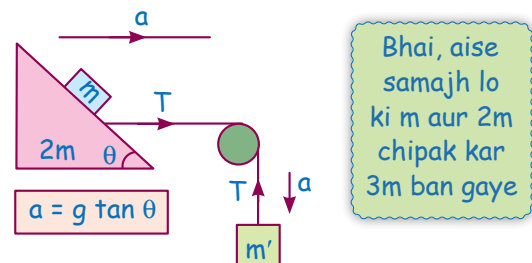


Now resolve mg and ma along the incline and perpendicular to incline.

Agar main wedge ke upar jaakar baith jaun toh mujhe block ruka hua dikhai dega, bas baithne se pehle pseudo force dena zaroori hai



Q. Find the value of m' so that m is at rest with respect to $2m$.



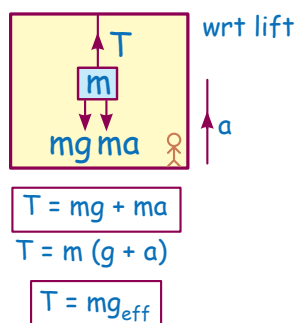
Sol. $a = g \tan \theta$

$$m'g - T = m'(g \tan \theta) \quad \dots(i)$$

$$T = (3m) g \tan \theta \quad \dots(ii)$$

Solving above equations we get m' .

★ Consider a lift moving upward with acc a , and a mass m is hanging inside lift by a string as shown in fig. Find tension.

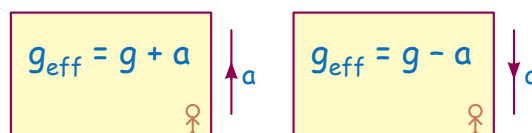


Ek kaam kar, chup chap lift ke andar jaakar baith ja. Baithne se pehle neeche pseudo force ma laga de. Block tujhe ruka hua dikhai dega, matlab $T = mg + ma$
 $\Rightarrow T = m(g + a)$
 $\Rightarrow T = mg_{\text{eff}}$

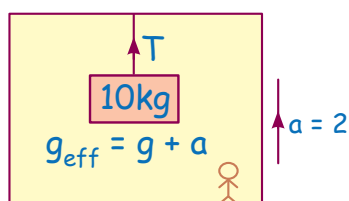


SKC

g_{eff} matlab aise samajh lo, lift ke andar baithne par aap aise duniya mein aagaye jahan g ki value $g \pm a$ hai.



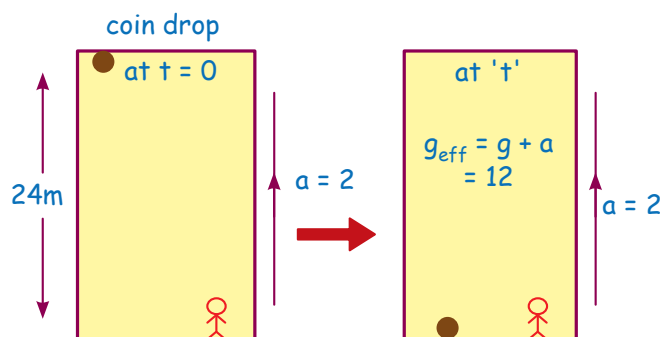
Q. Find tension in string.



Sol. $T = m g_{\text{eff}} = m (g + a)$

$T = 10 \times 12 = 120$

Q. A coin is dropped at $t = 0$ from ceiling of a lift as shown in fig. Find time when it will hit the floor of lift.



Sol. $g_{\text{eff}} = 12 \text{ m/s}^2$

$24 = 0 + \frac{1}{2} \times 12 \times t^2$

$t = 2$

$t = \sqrt{\frac{2h}{g_{\text{eff}}}}$

Pahle Kinematics

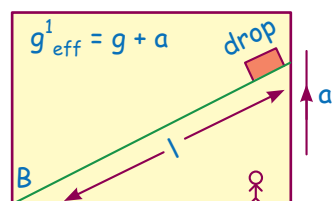
$U_{\text{coin/lift}} = 0, a_{\text{coin/lift}} = -10 - 2$

$S_{\text{coin/lift}} = -24 = -12$

$-24 = 0 + \frac{1}{2} (-12) t^2$

$t = 2 \text{ sec}$

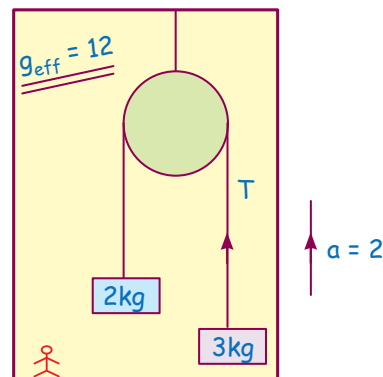
Q. Find time, when the block will reach at B.



Sol. $l = 0 + \frac{1}{2} \times (g + a) \sin \theta t^2$

$\Rightarrow t = \sqrt{\frac{2l}{(g + a) \sin \theta}}$

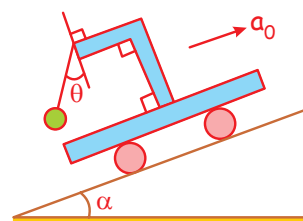
Q. Find tension in the string in following case.



Sol. $T = \frac{2m_1m_2}{m_1 + m_2} g_{\text{eff}} = \frac{2 \times 2 \times 3}{2 + 3} \times 12$



Q. A pendulum of mass m hangs from a support fixed to a trolley. The direction of the string when the trolley rolls up a plane of inclination α with acceleration a_0 is (String and bob remain fixed with respect to trolley)

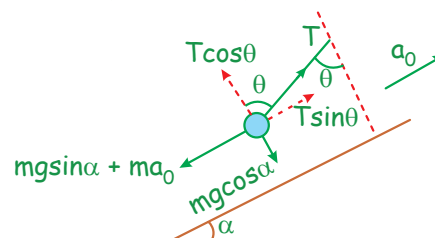


Sol. Balancing forces in the frame of trolley (non-inertial frame)

$T \sin \theta = m (g \sin \alpha + a_0)$

$T \cos \theta = mg \cos \alpha$

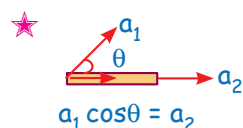
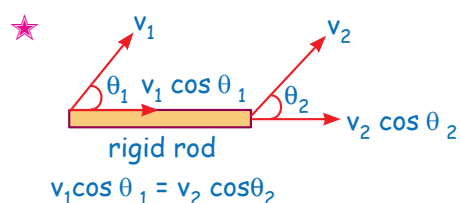
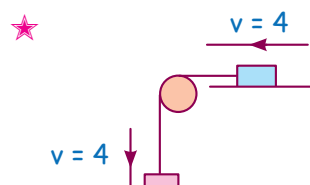
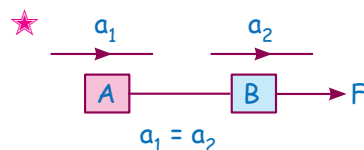
$\Rightarrow \tan \theta = \left(\frac{g \sin \alpha + a_0}{g \cos \alpha} \right)$



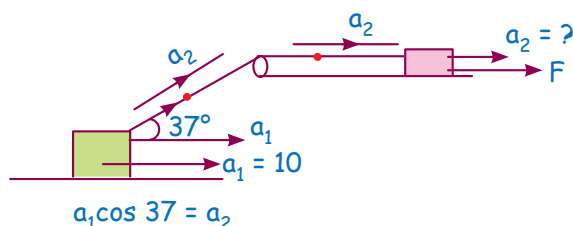
$\theta = \tan^{-1} \left(\frac{g \sin \alpha + a_0}{g \cos \alpha} \right)$

CONSTRAINT MOTION

Along the rigid rod or taut string; Components of magnitude of velocity & acc. should be same.



Q. Relation between a_1 and a_2 .



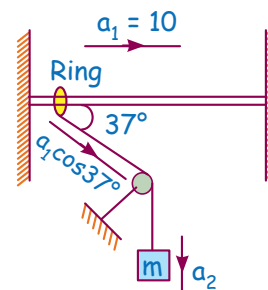
SKC

Dekh bhai, seedhi baat no bakwas, block ki actual velocity/acc ko rasssi/rod ki taraf tod lo aur barabar kar do.

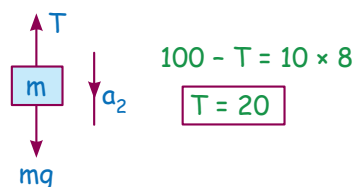


Q. Ring is constant to move along the fix rod as shown in fig. If at a instant acc of ring is 10 m/s^2 along +x-axis.

(a) Find acceleration of block.



(b) If $m = 10 \text{ kg}$ find T at this instant



Virtual Work Method

Ye kaise aaya, iska reason WPE (Work-Energy Theorem) mein padhenge (net Work Done by internal tension = 0). Abhi ye sikho isko apply kaise karna hai

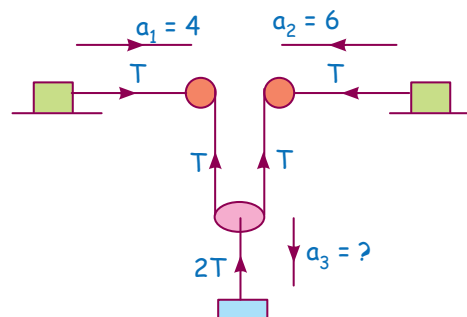
differentiate

$$\vec{T}_1 \cdot \vec{x}_1 + \vec{T}_2 \cdot \vec{x}_2 + \dots = 0$$

$$\vec{T}_1 \cdot \vec{v}_1 + \vec{T}_2 \cdot \vec{v}_2 + \dots = 0$$

$$\vec{T}_1 \cdot \vec{a}_1 + \vec{T}_2 \cdot \vec{a}_2 + \vec{T}_3 \cdot \vec{a}_3 + \dots = 0$$

Q. Find a_3 .

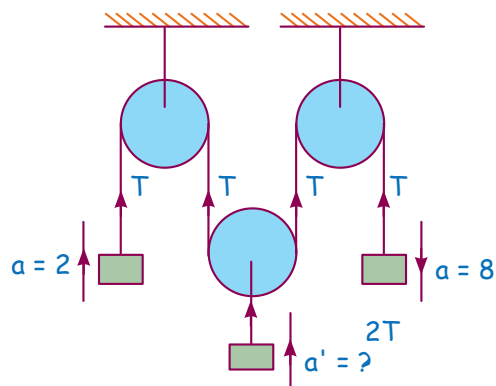


$$\vec{T}_1 \cdot \vec{a}_1 + \vec{T}_2 \cdot \vec{a}_2 + \vec{T}_3 \cdot \vec{a}_3 = 0$$

$$T \times 4 + T \times 6 - 2Ta_3 = 0$$

$$a_3 = 5$$

Q. Find a' in following figure.



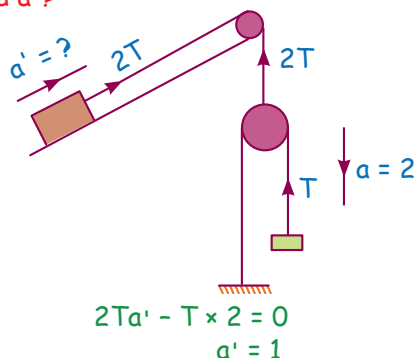
Sol. $T \times 2 + 2Ta' - T \times 8 = 0$
 $a' = 3$

Kuchh bachcho ke dimag mein ye doubt aayega ki sir yahan mass toh given nahi hai



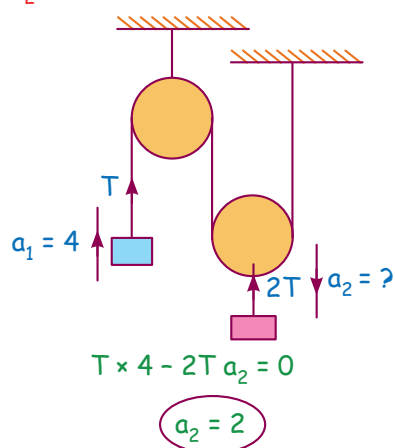
Beta yahan mass ki zaroorat nahi hai, yahi toh seekhna hai. Agar rassi mein tension puchhta toh mass ki zaroorat hoti

Q. Find a' ?



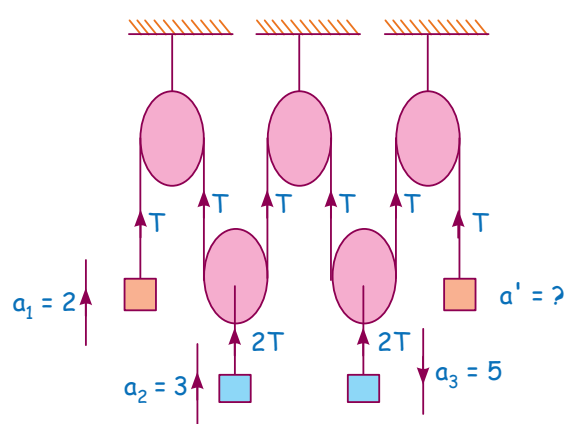
$2Ta' - T \times 2 = 0$
 $a' = 1$

Q. Find a_2 .



$T \times 4 - 2Ta_2 = 0$
 $a_2 = 2$

Q. Find a' in following figure.



$T \times 2 + 2T \times 3 - 2T \times 5 + Ta' = 0$
 $2 + 6 - 10 + a' = 0$

$a' = 2$

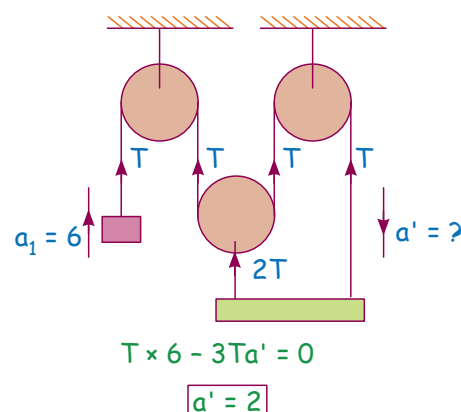
Dekh bhai, kisi ek rassi mein tension T maan aur uske baad sabhi rassi mein tension nikal de. Jiska acceleration nahi pata, uska acceleration upar ki taraf maan lo aur solve kar do. Agar answer negative aaya toh samajh jaana acceleration neeche tha



Abe ja kahan raha hai? Pehle aage wale ques aur solve kar

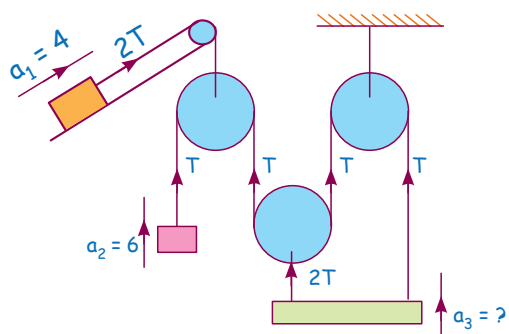


Q. Find a' in following figure.



$T \times 6 - 3Ta' = 0$
 $a' = 2$

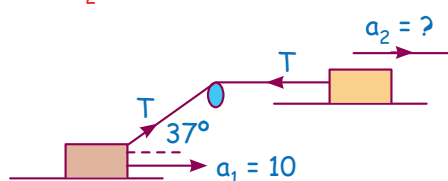
Q. Find a' in following figure.



$$4 \times 2 + 6 \times 1 + 3 \times a_3 = 0$$

$$a_3 = -\frac{14}{3} \text{ ('-' sign means downward acceleration)}$$

Q. Find a_2 ?



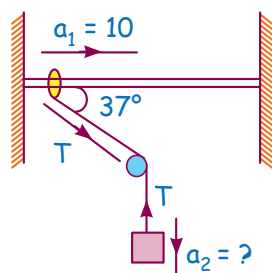
$$T_1 a_1 + T_2 a_2 = 0$$

$$T a_1 \cos 37^\circ + T a_2 \cos 180^\circ = 0$$

$$a_1 \cos 37^\circ - a_2 = 0$$

$$a_2 = a_1 \cos 37^\circ = 8$$

Q. Find a_2 ?

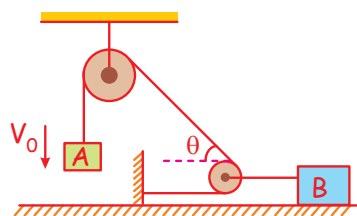


$$T a_1 \cos 37^\circ - T a_2 = 0$$

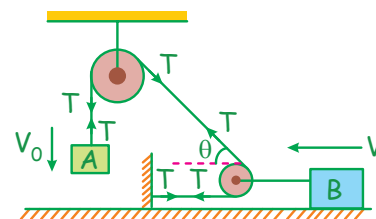
$$a_2 = a_1 \cos 37^\circ$$

$$a_2 = 8$$

Q. In the figure, find the velocity of block B, if velocity of A is V_0 in downward direction?

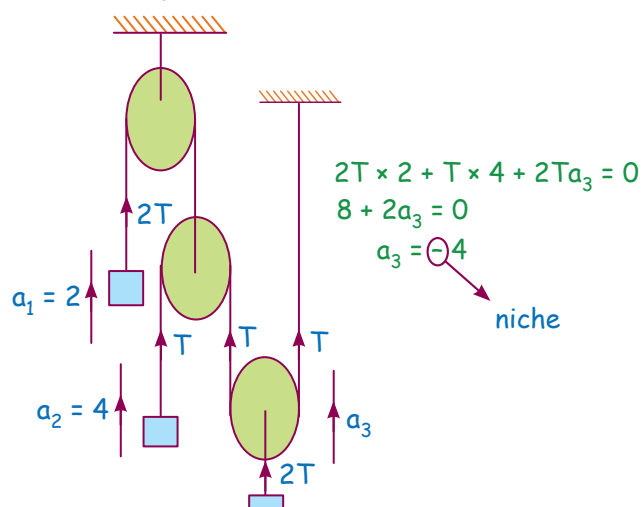


Sol.



$$-TV_0 + TV \cos \theta + TV = 0 \Rightarrow V = \frac{V_0}{1 + \cos \theta}$$

Q. Find a_3 .



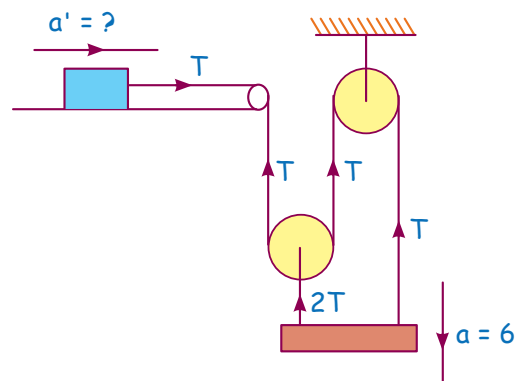
$$2T \times 2 + T \times 4 + 2Ta_3 = 0$$

$$8 + 2a_3 = 0$$

$$a_3 = -4$$

niche

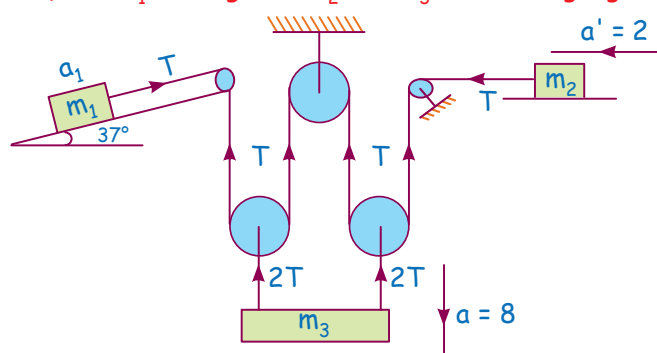
Q. Find a' and T .



$$Ta' - 3T \times 6 = 0$$

$$a' = 18$$

Q. If $m_1 = 10$ kg find m_2 and m_3 in following fig.



$$Ta_1 - 32T + 2T = 0$$

$$a_1 = 30$$

$$T - 100 \sin 37 = 10 \times 30 \text{ (For } m_1)$$

$$T = 360$$

$$m_3 g - 4T = m_3 \times a$$

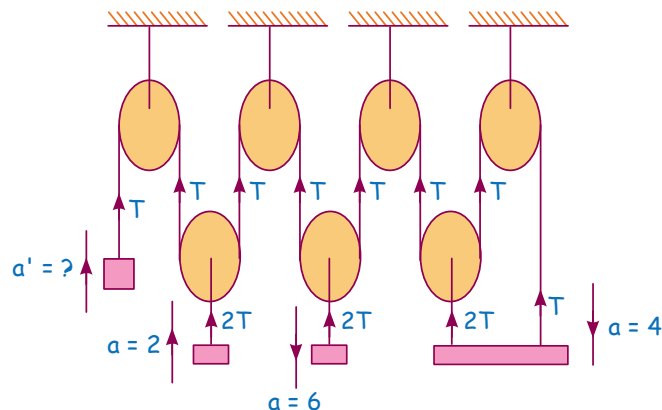
$$m_3 \times 10 - 4 \times 360 = m_3 \times 8$$

$$m_3 = 720 \text{ kg}$$

$$T = m_2 a' \Rightarrow 360 = m_2 \times 2$$

$$m_2 = 180$$

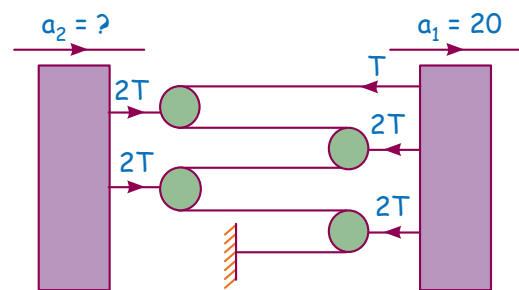
Q. Find a'



$$Ta' + 4T - 12T - 12T = 0$$

$$a' = 20$$

Q. Find a_2

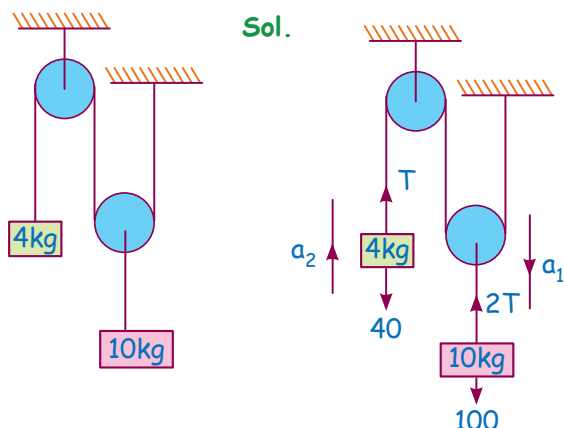


$$4Ta_2 - 5T \cdot 20 = 0$$

$$a_2 = 25$$

Q. Find acc of each block in following case.

Sol.



Method-1:

$$T - 40 = 4a_2 \quad \dots(1)$$

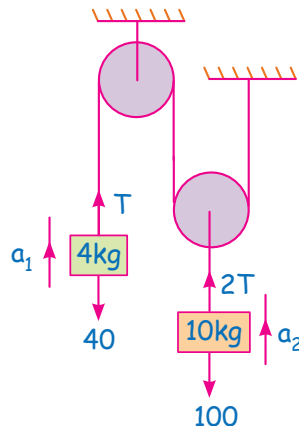
$$100 - 2T = 10a_1 \quad \dots(2)$$

$$Ta_2 - 2Ta_1 = 0$$

$$a_2 = 2a_1 \text{ (constraint relation)} \quad \dots(3)$$

Solve & get

Method-2: SKC Method



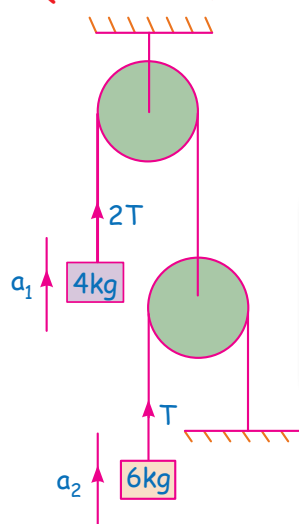
$$Ta_1 + 2Ta_2 = 0$$

$$a_1 + 2a_2 = 0$$

$$\frac{T - 40}{4} + 2 \left(\frac{2T - 100}{10} \right) = 0$$

Solve and get value of T and then a_1 and a_2

Q. Find acc of each block.



$$2a_1 + a_2 = 0$$

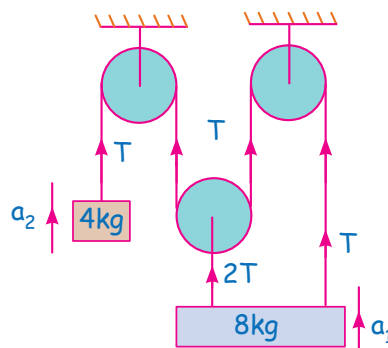
$$2 \left(\frac{2T - 40}{4} \right) + \frac{T - 60}{6} = 0$$

Now you can solve and get the value of tension.

Abe uske baad block ki equation likhkar a_1 a_2 nikaal loge na



Q. Find acc of each block.

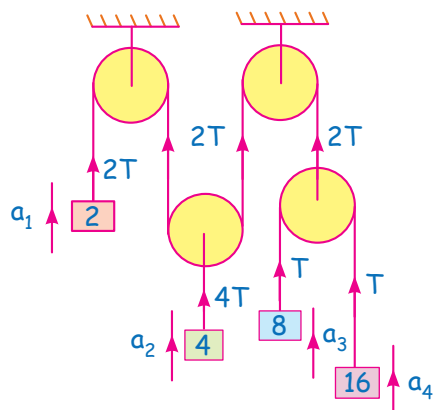


$$\text{Sol. } Ta_2 + 3Ta_1 = 0$$

$$a_2 + 3a_1 = 0$$

$$\frac{T - 40}{4} + 3 \left(\frac{3T - 80}{8} \right) = 0$$

Q. Find acc of each block.

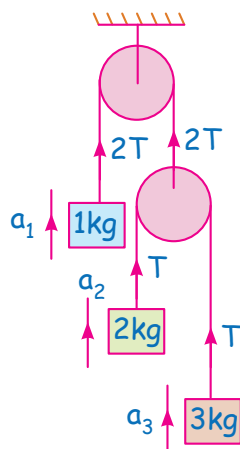


Sol. $2a_1 + 4a_2 + a_3 + a_4 = 0$

$$2\left(\frac{2T - 20}{2}\right) + 4\left(\frac{4T - 40}{4}\right) + \left(\frac{T - 80}{8}\right) + \frac{T - 160}{16} = 0$$

Ye zaroor check kar lena ki jitne mass, utne terms aayenge

Q. Find acc of each block.



Sol. $2Ta_1 + Ta_2 + Ta_3 = 0$

$$2a_1 + a_2 + a_3 = 0$$

$$2\left(\frac{2T - 10}{1}\right) + \frac{T - 20}{2} + \frac{T - 30}{3} = 0$$

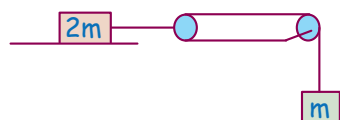
Solve and get T

$$a_1 = (2T - 10)/1$$

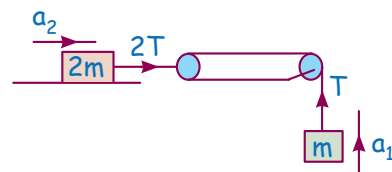
$$a_2 = (T - 20)/2$$

$$a_3 = (T - 30)/3$$

Q. Find acc of both the block.



Sol.



$$2Ta_2 + Ta_1 = 0$$

$$2a_2 + a_1 = 0$$

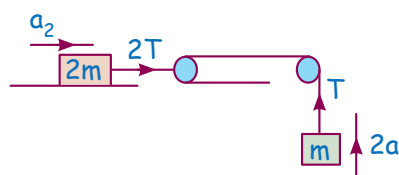
$$2\left(\frac{2T}{2m}\right) + \left(\frac{T - mg}{m}\right) = 0$$

$$2T + T - mg = 0$$

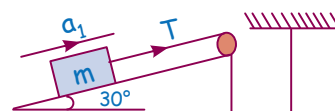
$$T = mg/3$$

$$a_1 = \frac{T - mg}{m} = \frac{mg/3 - mg}{m} = -\frac{2g}{3}$$

niche



Q. Find acc of both the block.

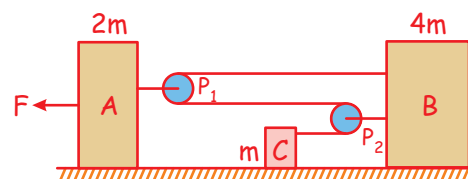


$$a_1 + 2a_2 = 0$$

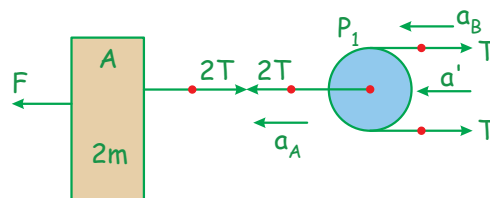
$$\frac{T - mg \sin 30}{m} + 2\left(\frac{2T - 2mg}{2m}\right) = 0$$

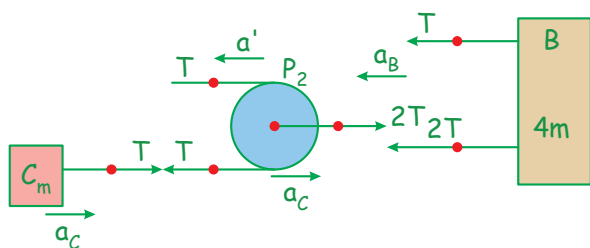
Now solve by yourself.

Q. In the shown figure, find the acceleration of block B. All surfaces are frictionless.



Sol.





$$-2Ta_A + 3Ta_B + Ta_C = 0 \Rightarrow 3a_B + a_C = 2a_A$$

From above equation

$$3a_B + a_C = 2a_A \quad \dots(i) \text{ [Constraint relation]}$$

$$F - 2T = 2ma_A \dots(ii) \text{ [Newton's II law for block A]}$$

$$3T = 4m a_B \dots(iii) \text{ [Newton's II law for block B]}$$

$$T = ma_C \quad \dots(iv) \text{ [Newton's II law for block C]}$$

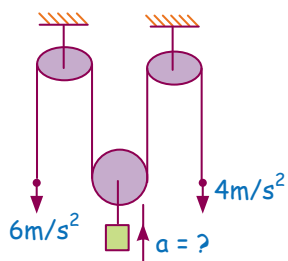
From (i), (ii), (iii) and (iv)

$$3 \times \frac{3T}{4m} + \frac{T}{m} = 2 \left(\frac{F - 2T}{2m} \right)$$

$$\frac{9T}{4} + T = F - 2T \Rightarrow T = \frac{4F}{21}$$

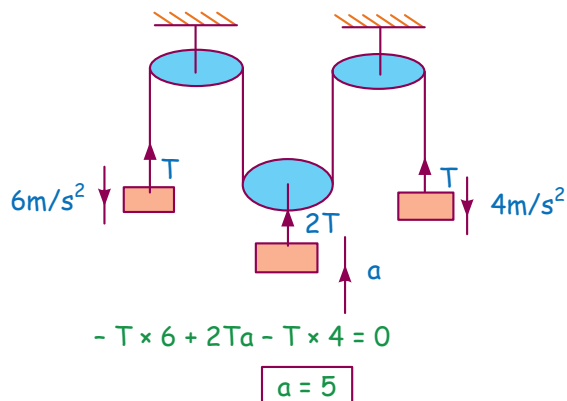
$$a_B = \frac{3T}{4m} \Rightarrow a_B = \frac{F}{7m}$$

Q. Find a ?

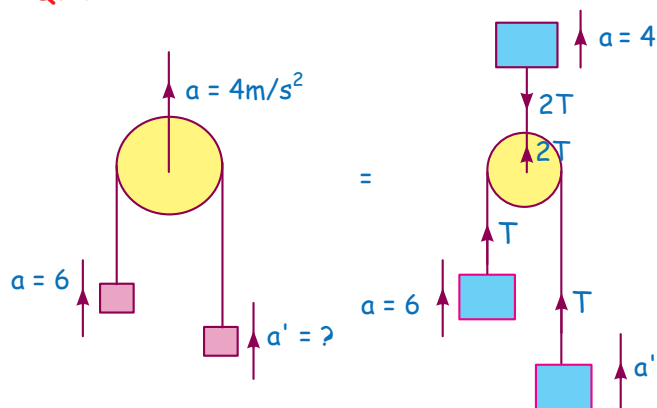


Is type ke questions mein agar block missing hai toh zaroorat ke hisab se bana lo

Sol.



Q. Find a' ?

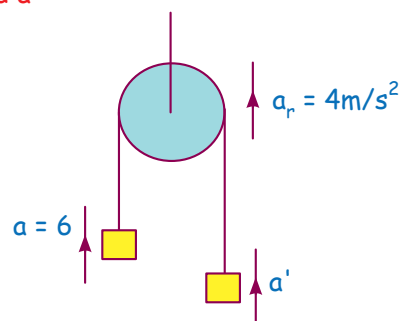


$$+T \times 6 + Ta' - 2T_4 = 0$$

$$+6 + a' - 8 = 0$$

$$a' = 2$$

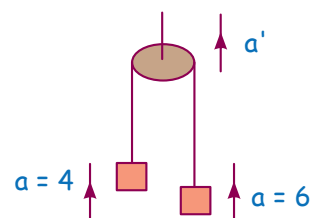
Q. Find a'



$$4 = \frac{6 + a'}{2} \text{ (Ye ek mast wala result hai. Kaise aaya wo chhodho, iska pattern yaad kar lo)}$$

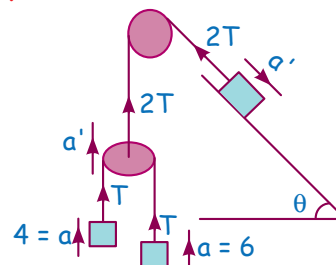
$$a' = 2$$

Q. Find a'



$$\text{Sol. } a' = \frac{4 + 6}{2} = 5$$

Q. Find a' .

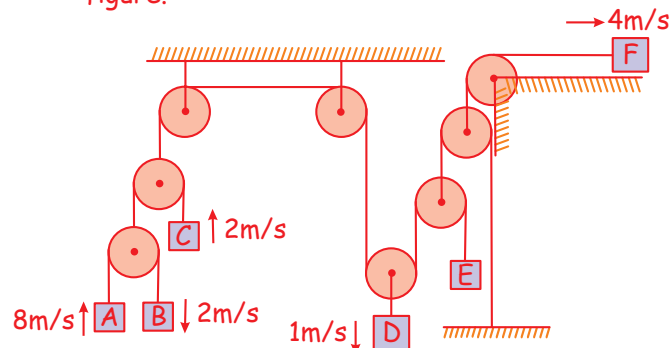


Method-1: $T \cdot 4 + T \cdot 6 - 2Ta' = 0$

$$a' = 5$$

Method-2: $a' = \frac{4+6}{2} = 5$

Q. Find out the velocity of block E as shown in figure.

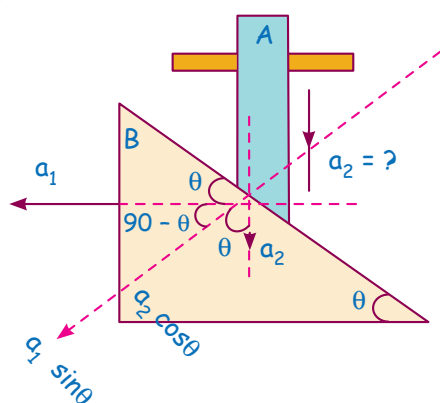


Sol. $8T - 2T + (2T \times 2) - (8T \times 1) + 4TV_E - (16T \times 4) = 0$

$$4V_E = -8 + 2 - 4 + 8 + 64 = 62$$

$$V_E = \frac{62}{4} = \frac{31}{2} \text{ ms}^{-1}$$

WEDGE CONSTRAINT



Method-1:

- ★ There no relative motion along common normal.
- ★ Component of acc of A & B along the common normal same

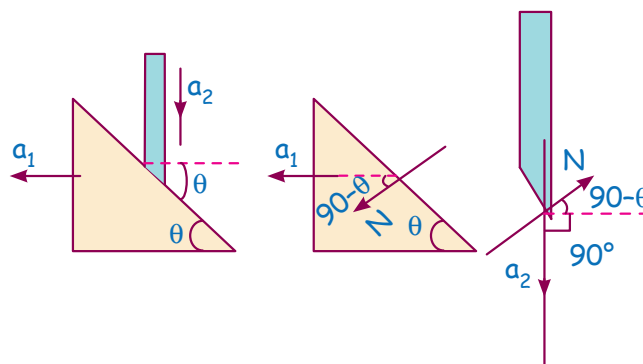
$$a_1 \sin \theta = a_2 \cos \theta$$

SKC

Dekh bhai, seedhi baat no bakwas,
dono bando ka Acc normal ki taraf
barabar kar do



Method-2:



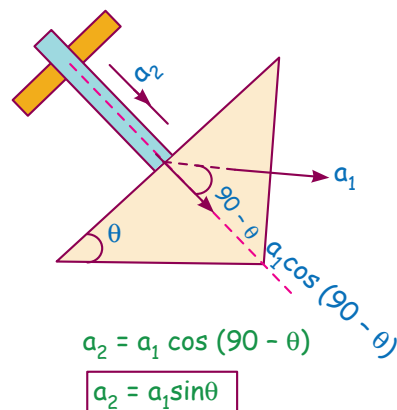
$$N_1 \cdot a_1 + N_2 \cdot a_2 = 0$$

$$Na_1 \cos (90 - \theta) + Na_2 \cos (90 + 90 - \theta) = 0$$

$$a_1 \sin \theta - a_2 \cos \theta = 0$$

$$a_1 \sin \theta = a_2 \cos \theta$$

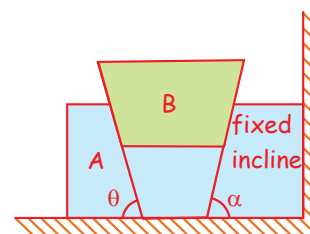
Q. Find constraint relation for following fig.



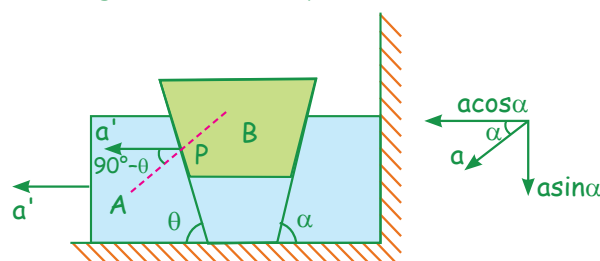
$$a_2 = a_1 \cos (90 - \theta)$$

$$a_2 = a_1 \sin \theta$$

Q. In the arrangement shown in figure, if the acceleration of B is a, then find the acceleration of A.



Sol. Let a' be the acceleration of block A using wedge constraint at point P

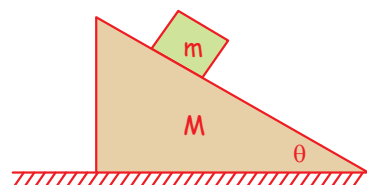


$$a \sin \alpha \cos \theta + a \cos \alpha \sin \theta = a' \cos(90^\circ - \theta)$$

$$a \sin(\alpha + \theta) = a' \sin \theta$$

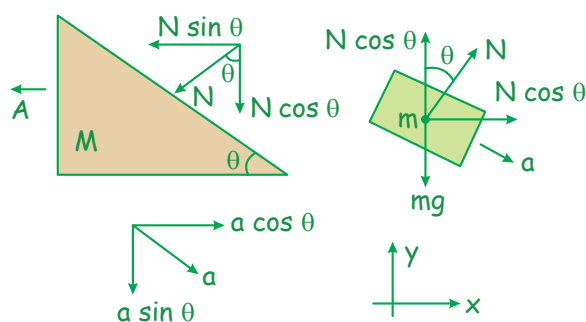
$$a' = \frac{a \sin(\theta + \alpha)}{\sin \theta}$$

Q. A block of mass m is placed on the inclined surface of a wedge as shown. Calculate the acceleration of the wedge and the block when the block is released. Assume all surfaces are frictionless.



Sol. Let the acceleration of wedge be A and that of block be a (w.r.t. wedge). Then acceleration of m w.r.t. ground is

$$(a \cos \theta - A)\hat{i} - a \sin \theta \hat{j}$$



$$\text{For } M: N \sin \theta = MA \quad \dots (i)$$

$$\text{For } m: mg - N \cos \theta = ma \sin \theta \quad \dots (ii)$$

$$N \sin \theta = m(a \cos \theta - A) \quad \dots (iii)$$

Solved (i), (ii) and (iii) to get

$$A = \frac{mg \sin \theta \cos \theta}{M + m \sin^2 \theta} \text{ and } a = \frac{(M + m)g \sin \theta}{M + m \sin^2 \theta}$$

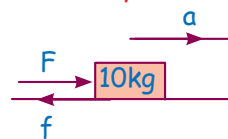
Now acceleration of block can be found by putting the values of A and a in (i).

FRICITION

Dekh bhai ye koi naya chapter nahi hai, NLM hi hai isme ek force padhenge jiska naam hai friction aur use purane forces ke saath mix karenge. Bas is baat ka dhyān rakhna ki har question me friction kitna lag raha hai aur uska nature kya hai



Q. A block of mass 10 kg is at rest on rough surface. A boy is applying force F on the block such that value of F is increasing slowly from zero analyse the situation



$$F - f = ma$$



F	a	f	
0	0	0	
2N	0	2 Niche	f badal rha h block is at rest (static friction)
5	0	5 "	
10	0	10 "	
20	0	20	
39.99	✓	39.99	Kinetic friction (f_k)
40.001	✓	30 (let)	
45	✓	30	
50	✓	30	
100	✓	30	

Static Friction

It is variable force and self adjusting force.

Experimentally $\Rightarrow (f_s)_{\max} \propto N$

$$(f_s)_{\max} = \mu_s N$$

coefficient of static friction

★ It opposes relative motion b/w contact surface

★ $(f_s)_{\max}$ = limiting friction

★ Independent on area of contact.

Kinetic Friction

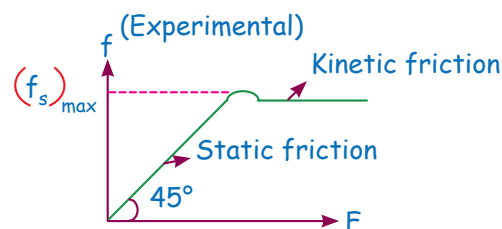
- It opposes the relative motion b/w contact surface

$$f_k \propto N$$

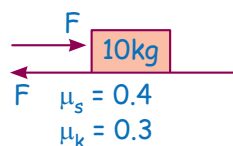
$$f_k = \mu_k N = \text{Const}$$

→ Coefficient of kinetic friction

★ It's value is constant and independent of area of contact surface.



Q. In given fig. Find value of acc and friction if $F = 10 \text{ N}, 20 \text{ N}, 35 \text{ N}, 39.99 \text{ N}, 40.01 \text{ N}, 50 \text{ N}$.



Sol. $(f_s)_{\max} = \mu_s N = 0.4 \times mg = 40 \text{ N}$

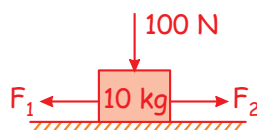
$$f_k = \mu_k N = 0.3 \times 100 = 30 \text{ N}$$



Dekh bhai sabse pehle carefully $(f_s)_{\max}$ aur f_k nikalo. Yahan $(f_s)_{\max} = 40 \text{ N}$ aaya, matlab ruke hue block par agar 40 N ya usse kam force lagaoge, toh block ruka rahega aur friction static lagega, jiski value jitna force lagaya uske barabar hogi. Agar 40 N se zyada force lagaya, toh block move kar jayega. $f_k = 30 \text{ N}$ aaya, means agar block move kar raha hai, toh friction kinetic lagega, jiski value constant 30 N hogi.

- (a) Find a & f if $F = 10 \text{ N}$
 $a = 0, f = 10 \text{ N}, (\text{Static})$
- (b) Find a & f if $F = 20 \text{ N} \Rightarrow a = 0, f_s = 20$
- (c) " " " " $F = 35 \text{ N} \Rightarrow a = 0, f_s = 35$
- (d) " " " " $F = 39.99 \Rightarrow a = 0, f_s = 39.99$
- (e) " " " " $F = 40.01 \Rightarrow a = \frac{F - f_k}{m}$
 $a = \frac{40.01 - 30}{10} = 1.001$
- (f) $F = 50 \text{ N}, a = \frac{50 - 30}{10} = 2$

Q. Find frictional force acting on the block in the following cases given that $\mu = 0.1$ [Take $g = 10 \text{ m/s}^2$]



Case-I: $F_1 = 0$

$$F_2 = 5 \text{ N}$$

$$F_2 = 15 \text{ N}$$

$$F_2 = 25 \text{ N}$$

Case-II: $F_1 = 20 \text{ N}$

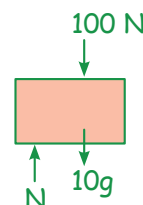
$$F_2 = 10 \text{ N}$$

$$F_2 = 20 \text{ N}$$

$$F_2 = 42 \text{ N}$$

$$F_2 = 50 \text{ N}$$

Sol. F.B.D. of the block



$$N = 10g + 100 = 200 \text{ N}$$

$$f_{s,\max} = \mu_s N = 200 \times 0.1 = 20 \text{ N}$$

If $|F_2 - F_1| < \mu_s N$ block will not move and self adjusting static friction will act on its block and will be equal to $|F_2 - F_1|$.

If $|F_2 - F_1| > \mu_s N$

then, $f_k = \mu_k N$

Case-I: $F_1 = 0$

$$F_2 = 5 \text{ N} \quad f = 5 \text{ N (towards left)}$$

$$F_2 = 15 \text{ N} \quad f = 15 \text{ N (towards left)}$$

$$F_2 = 25 \text{ N} \quad f = 20 \text{ N (towards left)}$$

Case-II: $F_1 = 20 \text{ N}$

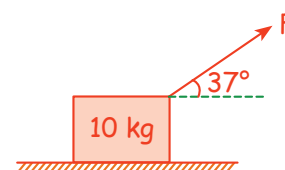
$$F_2 = 10 \text{ N} \quad f = 10 \text{ N (towards right)}$$

$$F_2 = 20 \text{ N} \quad f = 0$$

$$F_2 = 42 \text{ N} \quad f = 20 \text{ N (towards left)}$$

$$F_2 = 50 \text{ N} \quad f = 20 \text{ N (towards left)}$$

Q. A force F is applied on the block as shown in the figure. If the coefficient of static friction $\mu_s = 0.3$ and coefficient of kinetic friction is $\mu_k = 0.2$.

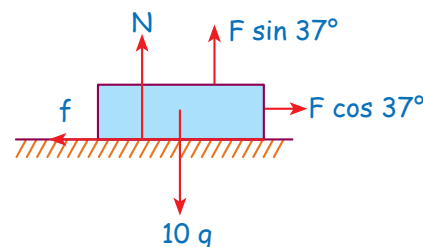


Find frictional force and acceleration of the block if

(i) $F = 10 \text{ N}$

(ii) $F = 50 \text{ N}$

Sol. F.B.D. of the block



(i) $F = 10 \text{ N}$

Balancing forces normal to the surface

$$N + 10\sin 37^\circ - 10g = 0 \Rightarrow N = 94 \text{ N}$$

$$f_{s, \max} = \mu_s N = 0.3 \times 94 = 28.2 \text{ N}$$

$$F\cos 37^\circ = 10 \times \frac{4}{5} = 8 \text{ N}$$

Since, $F\cos 37^\circ < \mu_s N$, block will not move and static friction $f_s = F\cos 37^\circ = 8 \text{ N}$.

(ii) $F = 50 \text{ N}$

$$N + 50\sin 37^\circ - 10g = 0 \Rightarrow N = 70 \text{ N}$$

$$f_{s, \max} = \mu_s N = 0.3 \times 70 = 21 \text{ N}$$

$$F\cos 37^\circ = 50 \times \frac{4}{5} = 40 \text{ N}$$

Since $F\cos 37^\circ > \mu_s N$, friction will not be able to prevent slipping.

Block will slide and kinetic friction will act on the block

$$f_k = 0.2 \text{ N} = 0.2 \times 70 = 14 \text{ N}$$

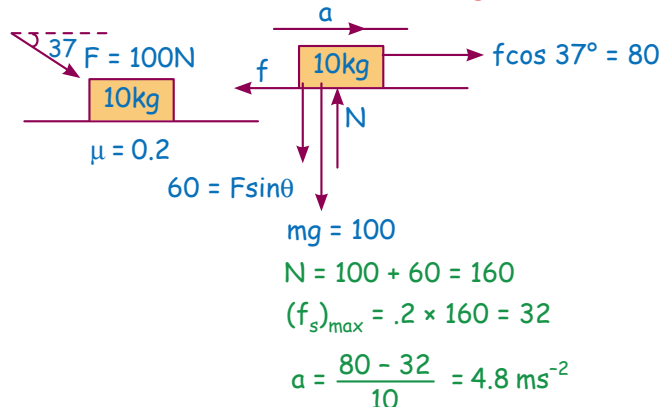
To find acceleration, we apply Newton's law along horizontal direction

$$F\cos 37^\circ - f_k = ma$$

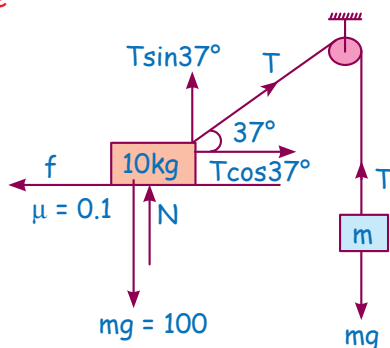
$$50\cos 37^\circ - 14 = 10a$$

$$\Rightarrow a = \frac{40 - 14}{10} = 2.6 \text{ m/s}^2$$

Q. Find the acc of block in following case.



Q. Find min value of m so that 10 kg block just move



$$T\sin 37^\circ + N = 100$$

$$\text{kayde me sahi} \Rightarrow T\cos 37^\circ > (f_s)_{\max}$$

$$T = mg$$

$$T\cos 37^\circ = (f_s)_{\max} = \mu N$$

$$mg \frac{4}{5} = 1 (100 - T\sin 37^\circ)$$

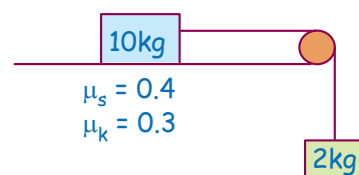
$$mg \frac{4}{5} = 10 - .1 \times mg \times \frac{3}{5}$$

$$mg \left(\frac{4}{5} \times \frac{3}{5} \right) = 10$$

$$m \times 10 \times \frac{4.3}{5} = 10$$

$$m = \frac{5}{4.3} \text{ kg}$$

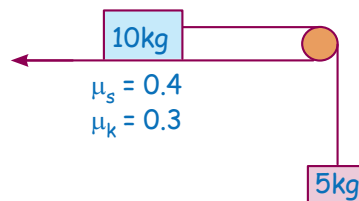
Q. Find acc. of block and value of friction.



Sol.

$$\begin{aligned} (f_s)_{\max} &= 40 \text{ N} \\ f_k &= 30 \text{ N} \\ a &= 0 \\ T &= 20 \\ f &= 20 \text{ (static)} \end{aligned}$$

Q. Find acc. of block and value of friction.



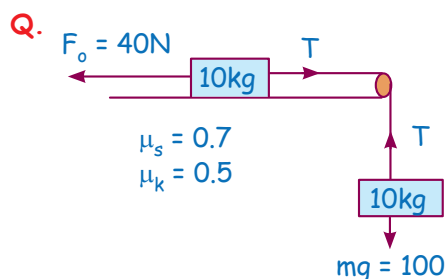
Sol. Since $(f_s)_{\max} = 40$ and $f_k = 30 \text{ N}$

$$mg = 50 > (f_s)_{\max} \text{ (Block will move)}$$

$$a = \frac{50 - f_k}{10 + 5} = \frac{50 - 30}{15} = \frac{4}{3}$$

$$50 - T = 5 \times a$$

$$T = 50 - 5 \times \frac{4}{3} = \frac{130}{3}$$



Sol.

SKC

Dekh bhai 10 kg ke block ko move karane ke liye effectively 70 N ka force chahiye aur yahi dhyan se dekho toh $100 - 40 = 60$ hai means block move nahi karega.

$$a = 0$$

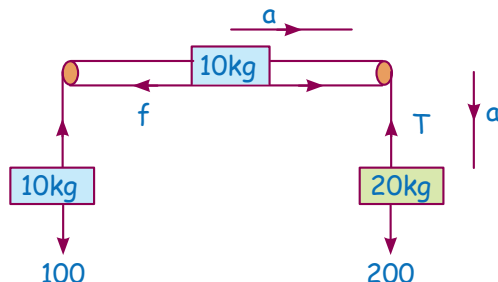
$$T = 100$$



$$T = F_o + f_s$$

$$100 = 40 + f_s$$

Q. Find the acc. of each block.



$$\mu_s = .6$$

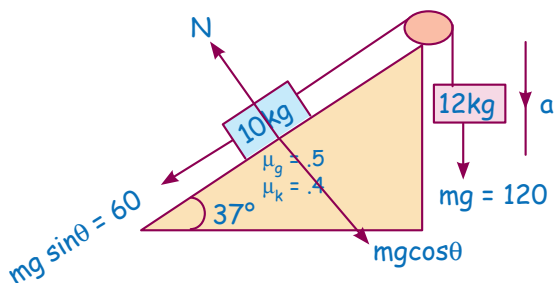
$$\mu_k = .4$$

$$(f_s)_{\max} = 60$$

$$f_k = 40$$

$$a = \frac{200 - 100 - 40}{10 + 10 + 20} = \frac{3}{2} \text{ ms}^{-2}$$

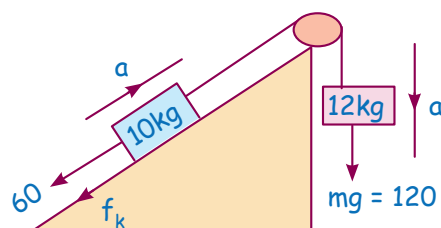
Q. Find acc. and friction.



$$N = mg \cos \theta = 80$$

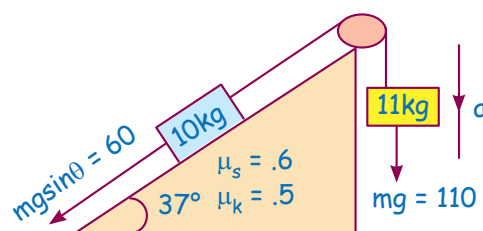
$$(f_s)_{\max} = .5 \times 80 = 40$$

$$f_k = .4 \times 80 = 32$$



$$a = \frac{120 - 60 - 32}{10 + 12}$$

Q. Find a?



$$(f_s)_{\max} = \mu_s N = \mu_s mg \cos \theta$$

$$(f_s)_{\max} = .6 \times 80 = 48$$

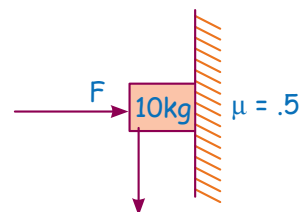
$$(f_k) = \mu_k N = .5 \times 80 = 40$$

$$a = \frac{110 - 60 - 40}{10 + 11} = \frac{10}{21} \text{ ms}^{-2}$$

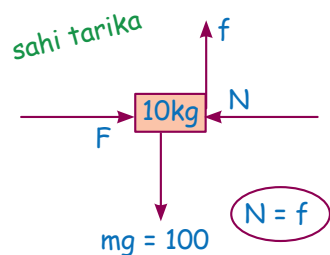
Friction ki understanding ko badhane ke liye neeche ques attach kar raha hoon jisse no. of page badhenge, publisher wale bhaiya gussa honge kyunki price toh hum same rakhenge lekin tumhe har ques solve karna hai. Bcz exam me in ques ke aane ki probability bahut zyada hai.



Q.



(a) Find min value of F so that block does not fall.



Block does not fall

$$mg \leq (f_s)_{\max}$$

$$mg \leq \mu_s N$$

$$100 \leq .5F$$

$$F \geq 200$$

$$F_{\min} = 200$$

Shortcut

$$mg = (f_s)_{\max} = \mu_s N$$

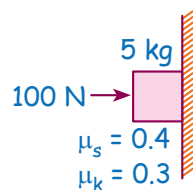
$$100 = 5 \times F_{\min}$$

$$F = 200 = F_{\min}$$

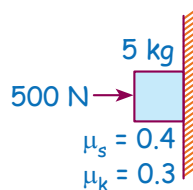
- (b) Find max value of F so that block fall
200 se kam lgaoge toh girega 199.99999.....

Sol. 200

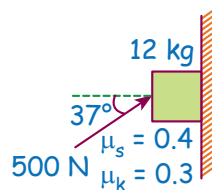
- Q.** Determine the magnitude of frictional force and acceleration of the block in each of the following cases: [Take $g = 10 \text{ m/s}^2$]



Case I

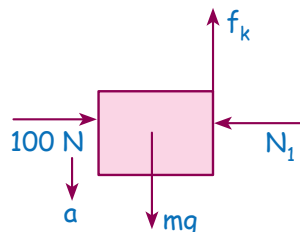


Case II



Case III

Sol. **Case-I:** $N_1 = 100 \text{ N}$, $mg = 50 \text{ N}$



$$f_1 = \mu_s N_1 = 0.4 \times 100 = 40 \text{ N}, f_k = \mu_k N_1$$

$$= 0.3 \times 100 = 30 \text{ N}$$

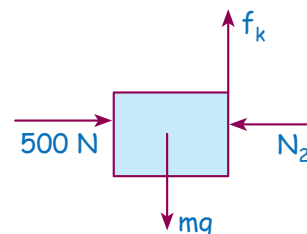
Here mg (driving force) is greater than maximum friction $f_1 = 40 \text{ N}$. Hence the block will not be able to stay at rest. It will accelerate downwards. But when it starts slipping, then kinetic friction will come into play.

Now

$$a = \frac{mg - f_k}{m} = \frac{50 - 30}{5} = 4 \text{ ms}^{-2}$$

So in this case $f = f_k = 30 \text{ N}$ and $a = 4 \text{ ms}^{-2}$ (downwards)

Case-II: $N_2 = 500 \text{ N}$, $mg = 50 \text{ N}$

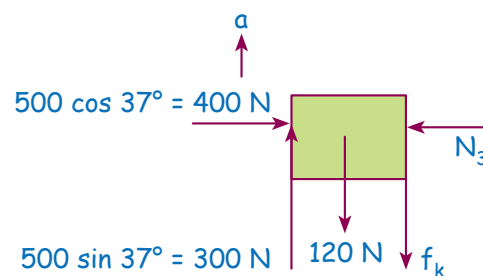


$$f_1 = \mu_s N_2 = 200 \text{ N}, f_k = \mu_k N_2 = 150 \text{ N}$$

Here f_1 is greater than mg (driving force), Hence, block will not move. So in this case $a = 0$, $f = mg = 50 \text{ N}$

Case-III:

$$N_3 = 400 \text{ N}, f_1 = \mu_s N_3 = 160 \text{ N}, f_k = \mu_k N_3 = 120 \text{ N}$$

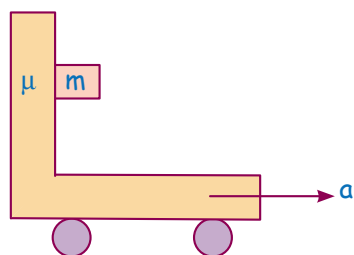


Here driving force $= 300 - 120 = 180 \text{ N}$ in upward direction; hence, friction will act downwards. Driving force is more than f_1 . So the block will accelerate upwards.

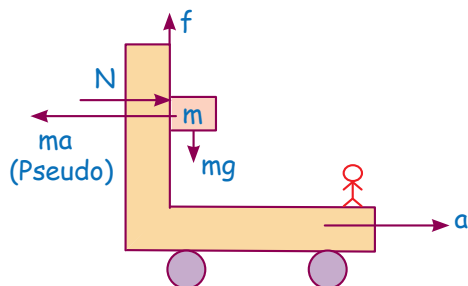
$$a = \frac{180 - f_k}{m} = \frac{180 - 120}{12} = 5 \text{ ms}^{-2} \text{ (upwards)}$$

So in this case $f = f_k = 120 \text{ N}$ and $a = 5 \text{ ms}^{-2}$ (upwards)

- Q.** (a) Find min acc 'a' so that block does not fall
(b) Find max acc 'a' so that block does fall



Sol.



सही $\Rightarrow mg \leq (f_s)_{\max}$

$$mg \leq \mu N$$

$$mg \leq \mu ma$$

$$\boxed{\frac{g}{\mu} \leq a}$$

$$N = ma \quad \dots(1)$$

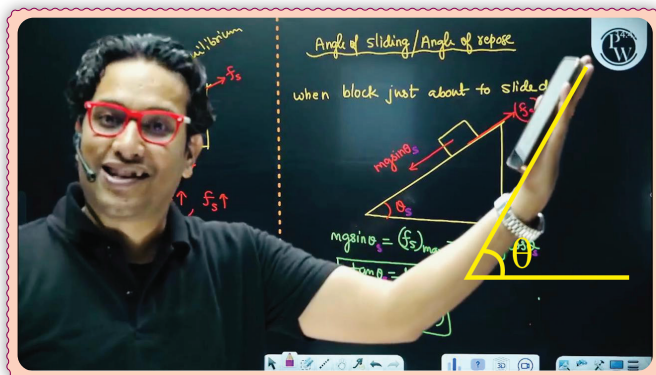
$$mg = (f_s)_{\max} = \mu N$$

$$mg = \mu ma$$

$$\boxed{a = \frac{g}{\mu}}$$

Answer for both the (a) and (b) is same and equal to $\frac{g}{\mu}$

Angle of Repose/Angle of Sliding

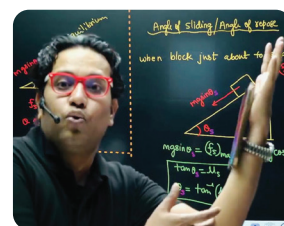
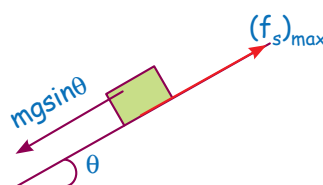


Agar main upar wali image me apne haath ko upar uthata jao, matlab θ increase karta jao, toh jis angle par ye mobile just neeche fisalna start ho jaega us angle ko hum fisalne wala angle kehte hai, bole toh angle of sliding



A block of mass m is at rest on a rough incline plane making angle θ with horizontal. The angle θ at wedge block is just about to slide down is called angle of repose.

When block is just about to slide



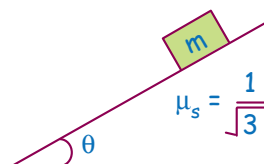
$$mg \sin \theta = (f_s)_{\max} = \mu_s mg \cos \theta$$

$$\mu_s = \tan \theta$$

$$\theta = \tan^{-1}(\mu_s) = \text{angle of repose}$$

$$\text{If } \theta \leq \theta_s = \text{Rest}$$

$$\theta > \theta_s = \text{Slide down}$$



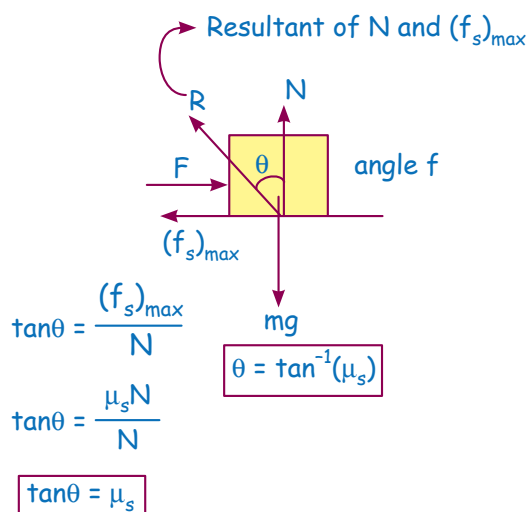
$$\tan \theta = \mu_s = \frac{1}{\sqrt{3}}$$

$$\theta_s = 30$$

$$\theta \leq 30 \Rightarrow \text{Block Rest}$$

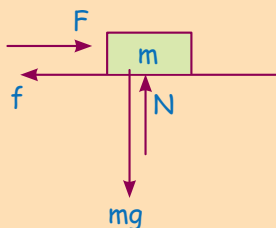
$$\theta > 30 \Rightarrow \text{Block Move}$$

Angle of Friction

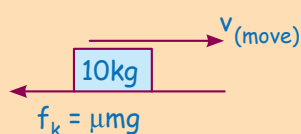




Kaam ka Dabba



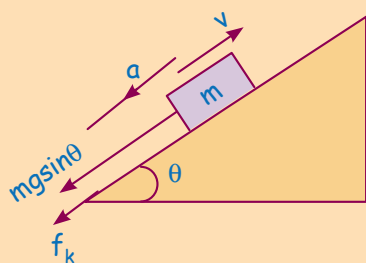
$$\text{Net contact force} = \sqrt{N^2 + f^2}$$



$$a = \frac{f_k}{m} = \frac{\mu mg}{m} = \mu g \text{ (piche)}$$

Stopping distance nikalne k liye 3rd eqn of motion lgao $0^2 = v^2 - 2(\mu g)x$

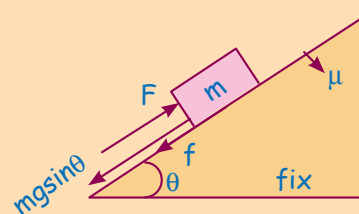
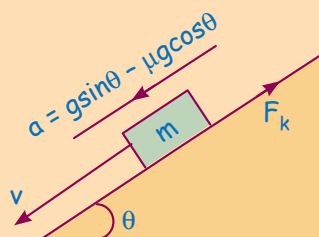
★ When block is moving up along the inclined



$$ma = mgsin\theta + \mu_k mgcos\theta$$

$$a = gsin\theta + \mu gcos\theta \text{ piche}$$

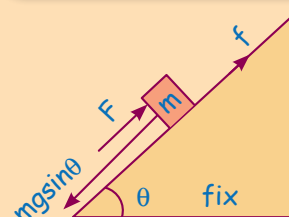
★ When block is moving down along the inclined



min value of F so that block slide up.

$$F_{\min} = mgsin\theta + (f_s)_{\max}$$

$$F_{\min} = mgsin\theta + \mu_s mgcos\theta.$$



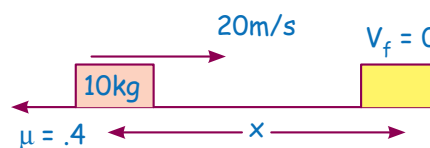
min value of F to prevent the block sliding down

$$F + (f_s)_{\max} = mgsin\theta$$

$$F = mgsin\theta - \mu mgcos\theta$$

Q. In following fig. Find distance travel by block before coming to rest.

Sol.

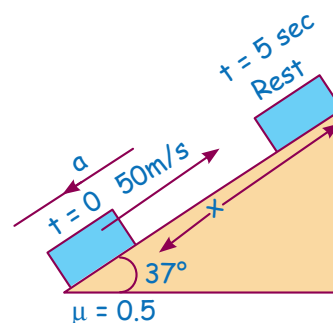


$$a = \mu g = 4 \text{ (piche)} = \text{const.}$$

$$0^2 = (20)^2 - 2 \times 4 \times x$$

$$x = 50 \text{ m}$$

Q. In the given fig. block is projected along the rough incline ($\mu = 0.5$) with speed 50 m/s. Find distance travel by block before coming to rest.



$$a = g \sin \theta + \mu g \cos \theta = 6 + .5 \times 8 = 10 \text{ m/s}^2$$

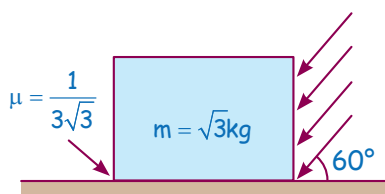
$$0^2 = (50)^2 - 2 \times 10 \times x$$

$$x = \frac{2500}{20} = 125 \text{ m}$$

Advance me directly friction se sawal nahi puche gaye hai, lekin mains me har saal repetative concept par question puche ja rahe hai. So, mai kuch important PYQ attach kar raha hoon inhe ache se solve kare, dekhna yahan se sawal zarur aayega.



- Q.** As shown in the figure, a block of mass $\sqrt{3} \text{ kg}$ is kept on a horizontal rough surface of coefficient of friction $\frac{1}{3\sqrt{3}}$. The critical force to be applied on the vertical surface as shown at an angle 60° with horizontal such that it does not move. The value of force will be ?



$$\left[g = 10 \text{ m/s}^2; \sin 60^\circ = \frac{\sqrt{3}}{2}; \cos 60^\circ = \frac{1}{2} \right]$$

Sol. Net force in the Vertical direction is zero so.

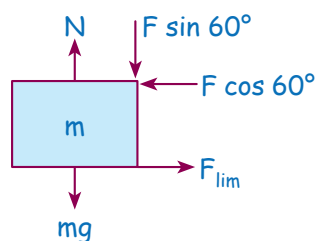
$$N = mg + F \sin 60^\circ$$

$$f_{\text{lim}} = F \cos 60^\circ \Rightarrow \mu N = F \cos 60^\circ$$

$$\Rightarrow \mu (mg + F \sin 60^\circ) = F \cos 60^\circ$$

$$\Rightarrow \mu mg = F [\cos 60^\circ - \mu \sin 60^\circ]$$

$$F = \frac{\mu mg}{\cos 60^\circ - \mu \sin 60^\circ} = \frac{\frac{1}{3\sqrt{3}} \times \sqrt{3} \times 10}{\frac{1}{2} - \frac{1}{3\sqrt{3}} \times \frac{\sqrt{3}}{2}} = 10$$

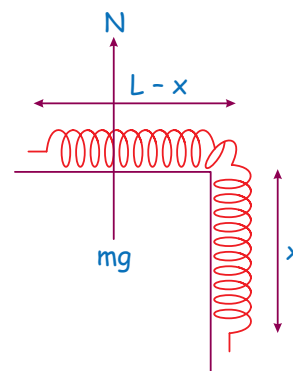


- Q.** A uniform chain of 6 m length is placed on a table such that a part of its length is hanging over the edge of the table. The system is at rest. The co-efficient of static friction between the chain and the surface of the table is 0.5, the maximum length of the chain hanging from the table is _____ m.

Sol. Mass per unit length = λ

$$N = mg = \lambda(L - x)g$$

$$f_{s, \text{max}} = \mu_s N$$



$$f_{s, \text{max}} = (0.5) (\lambda) (L - x)g$$

$$\text{And also } f_{s, \text{max}} = m_x g$$

$$0.5 \lambda (L - x)g = \lambda x g$$

$$\frac{L - x}{2} = x$$

$$\frac{L}{2} = \frac{3x}{2} \Rightarrow x = \frac{L}{3} = \frac{6}{3} = 2 \text{ m}$$

- Q.** A body of mass 1 kg rests on a horizontal floor with which it has a coefficient of static friction $\frac{1}{\sqrt{3}}$. It is desired to make the body move by applying the minimum possible force F (in N). The value of F will be _____ (Round off to the Nearest Integer) [Take $g = 10 \text{ ms}^{-2}$]

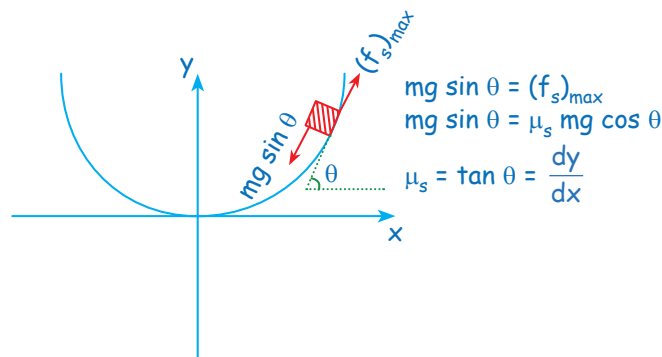
Sol. Minimum possible force,

$$F = \frac{\mu mg}{\sqrt{1 + \mu^2}}$$

$$F_{\text{min}} = \frac{\frac{1}{\sqrt{3}} \times 1 \times 10}{\sqrt{1 + \frac{1}{3}}} = 5 \text{ N}$$

- Q.** A block of mass m is placed on a surface having vertical cross section given by $y = x^2/4$. If coefficient of friction is 0.5, the maximum height above the ground at which block can be placed without slipping is:

Sol.

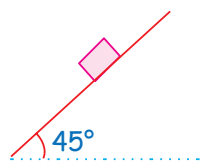


$$y = x^2/4, \mu = 0.5$$

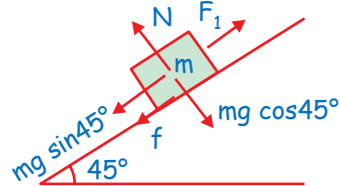
$$\frac{dy}{dx} = \tan \theta = \frac{x}{2} = \mu = \frac{1}{2}$$

$$x = 1, y = 1/4$$

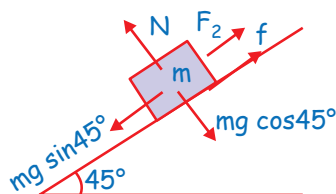
- Q.** Consider a block kept on an inclined plane (inclined at 45°) as shown in the figure. If the force required to just push it up the incline is 2 times the force required to just prevent it from sliding down, the coefficient of friction between the block and inclined plane (μ) is equal to:



- Sol.** (a) Force F_1 required to just push the block up the incline



$$F_1 = mg \sin 45^\circ + f = \frac{mg}{\sqrt{2}} + \mu mg \cos 45^\circ = \frac{mg}{\sqrt{2}}(1 + \mu)$$



Force F_2 required to prevent the block from sliding

$$F_2 = mg \sin 45^\circ - f$$

$$= mg \sin 45^\circ - \mu N = \frac{mg}{\sqrt{2}}(1 - \mu)$$

$$\text{Given, } F_1 = 2F_2$$

$$\Rightarrow \frac{mg}{\sqrt{2}}(1 + \mu) = 2 \frac{mg}{\sqrt{2}}(1 - \mu)$$

$$\Rightarrow \mu = 1/3 = 0.33$$

- Q.** The time taken by an object to slide down a 45° rough inclined plane is n times as it takes to slide down a perfectly smooth 45° inclined plane. The coefficient of kinetic friction between the object and the incline plane is

Sol. For a perfectly smooth inclined plane

$$a_1 = g \sin \theta = g / \sqrt{2}$$

For the rough inclined plane

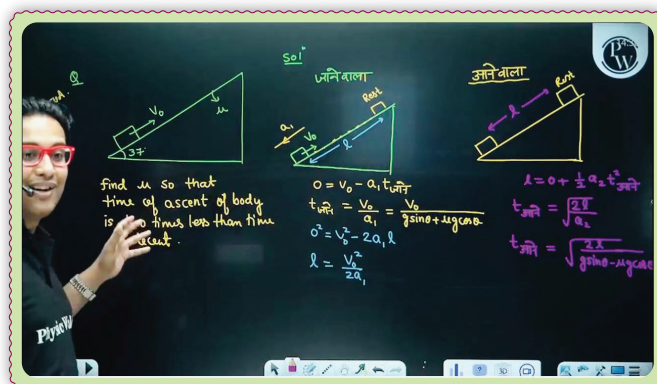
$$a_2 = g \sin \theta - \mu_k g \cos \theta = \frac{g}{\sqrt{2}} - \frac{kg}{\sqrt{2}}$$

$$\text{Given, } t_2 = nt_1$$

$$\text{and } a_1 t_1^2 = a_2 t_2^2$$

$$\Rightarrow \frac{g}{\sqrt{2}} t_1^2 = \left(\frac{g}{\sqrt{2}} - \frac{kg}{\sqrt{2}} \right) n^2 t_1^2$$

$$\Rightarrow \mu_k = 1 - \frac{1}{n^2}$$



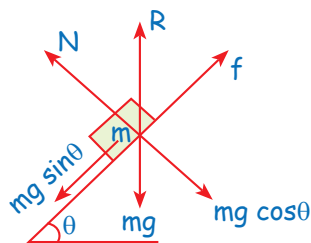
- Q.** A block of mass M slides down on a rough inclined plane with constant velocity. The angle made by the incline plane with horizontal is θ . The magnitude of the contact force will be:

Sol. From F. B. D

$$N = Mg \cos \theta$$

$$f = Mg \sin \theta$$

$$\therefore R = \sqrt{N^2 + f^2} = \sqrt{(Mg \cos \theta)^2 + (Mg \sin \theta)^2} = Mg$$



Q. A body of mass 'm' is launched up on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of friction between the body and plane is $\frac{\sqrt{x}}{5}$ if the time of ascent is half of the time of descent. The value of x is _____.

Sol. For Ascent:

$$t_a = \sqrt{\frac{2\ell}{a_{\text{ascent}}}} = \sqrt{\frac{2\ell}{g(\sin \theta + \mu \cos \theta)}} \quad \dots(i)$$

For Descent:

$$t_d = \sqrt{\frac{2\ell}{a_{\text{descent}}}} = \sqrt{\frac{2\ell}{g(\sin \theta - \mu \cos \theta)}} \quad \dots(ii)$$

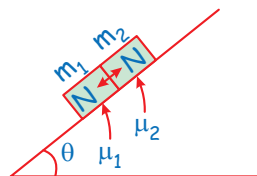
According to question, $t_a = \frac{1}{2} t_d$

$$\Rightarrow \frac{1}{\sin \theta + \mu \cos \theta} = \left(\frac{1}{4}\right) \left(\frac{1}{\sin \theta - \mu \cos \theta}\right) \quad [\text{From (i) and (ii)}]$$

$$\Rightarrow \mu = \frac{3}{5} \tan \theta = \frac{3}{5} \tan 30^\circ = \frac{\sqrt{3}}{5}$$

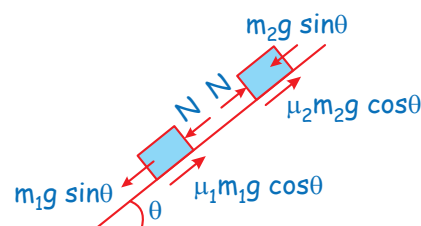
TWO BLOCKS ON AN INCLINED PLANE

Consider two blocks having masses m_1 and m_2 placed on a rough inclined plane. μ_1 and μ_2 are the friction coefficient for m_1 and m_2 respectively. If N is the normal force between the contact surface of m_1 and m_2 .



Now two condition arises.

(i) If $\mu_1 > \mu_2$ both will move with same acc



$$a = \frac{(m_1 + m_2)g \sin \theta - (\mu_1 m_1 + \mu_2 m_2)g \cos \theta}{m_1 + m_2}$$

We can also find normal force between block just by writing NLM eqn of any block.

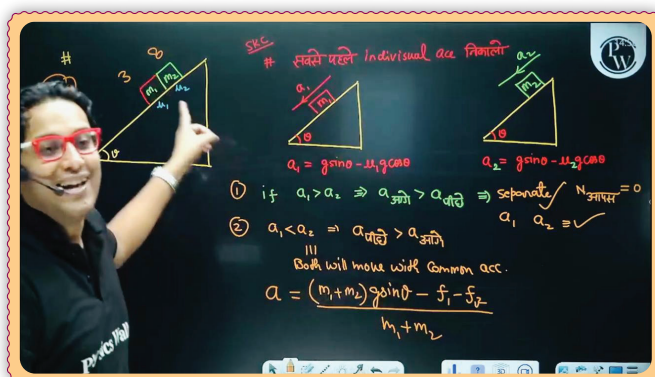
(ii) If $\mu_1 < \mu_2$ then, blocks will move with different acceleration.

$N = 0$ because, there is no contact between the blocks.

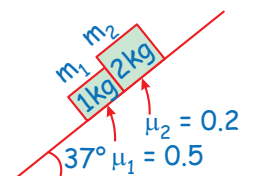
$$a_1 = g \sin \theta - \mu_1 g \cos \theta$$

$$a_2 = g \sin \theta - \mu_2 g \cos \theta$$

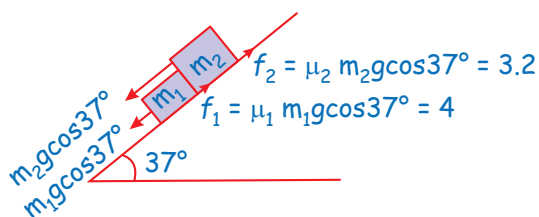
$$\Rightarrow a_1 > a_2$$



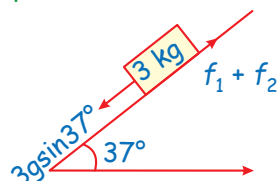
Q. Mass m_1 and m_2 are placed on a rough inclined plane as shown in figure. Find out the acceleration of the blocks and contact force in between these surface.



Sol. As we know if $\mu_1 > \mu_2$ both will travel together so $a_1 = a_2 = a$
F.B.D

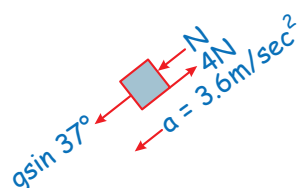


which is equivalent to



$$a = \frac{3g \sin 37^\circ - (f_1 + f_2)}{3} \Rightarrow a = \frac{18 - 7.2}{3} = 3.6 \text{ m/sec}^2$$

Now F.B.D of 1 kg block is

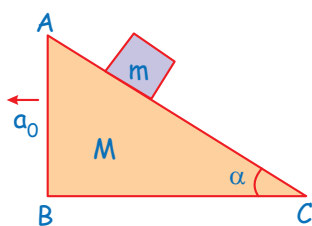


$$g \sin 37^\circ + N - 4 = (1) a$$

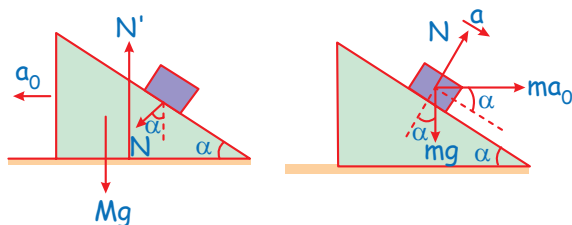
$$N = 3.6 + 4 - 6 = 1.6 \text{ Newton}$$

Now practice following question for Adv.

Q. All the surfaces shown in figure are assumed to be frictionless. The block of mass m slides on the prism which in turn slides backward on the horizontal surface. Find the acceleration of the smaller block with respect to the prism.



Sol. Let the acceleration of the prism be a_0 in the backward direction. Consider the motion of the smaller block from the frame of the prism. The forces on the block are (figure)



$$ma_0 \cos \alpha + mg \sin \alpha = ma \text{ (for block)}$$

$$\text{or; } a = a_0 \cos \alpha + g \sin \alpha \quad \dots (i)$$

$$N + ma_0 \sin \alpha = mg \cos \alpha \text{ (for block)} \quad \dots (ii)$$

$$N \sin \alpha = Ma_0 \text{ (for prism)}$$

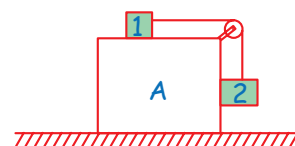
$$N = Ma_0 / \sin \alpha \quad \dots (iii)$$

Putting in (ii)

$$\frac{Ma_0}{\sin \alpha} + ma_0 \sin \alpha = mg \cos \alpha \text{ or } a_0 = \frac{mg \sin \alpha \cos \alpha}{M + m \sin^2 \alpha}$$

$$\text{From (i), } a = \frac{mg \sin \alpha \cos^2 \alpha}{M + m \sin^2 \alpha} + g \sin \alpha = \frac{(M + m) g \sin \alpha}{M + m \sin^2 \alpha}$$

Q. What is the minimum acceleration with which bar A should be shifted horizontally to keep the bodies 1 and 2 stationary relative to the bar? The masses of the bodies are equal and the coefficient of friction between the bar and the bodies equal to μ . The masses of the pulley and the threads are negligible while the friction in the pulley is absent.



Sol. Let us place the observer on A.

Since we have non-inertial frame we have pseudo forces.

For body '1' we have,

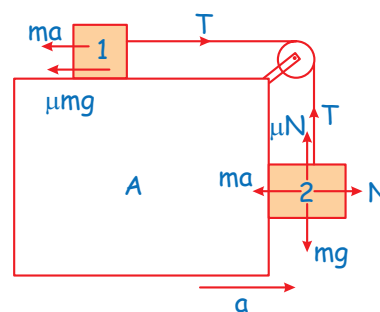
$$T = ma + \mu mg \quad \dots (i)$$

For body '2' we have,

$$N = ma$$

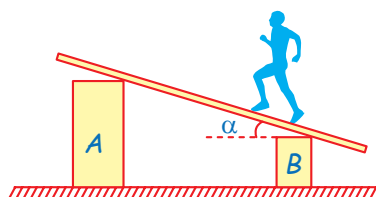
$$mg - T - \mu ma = 0$$

$$\therefore mg = T + \mu ma \quad \dots (ii)$$

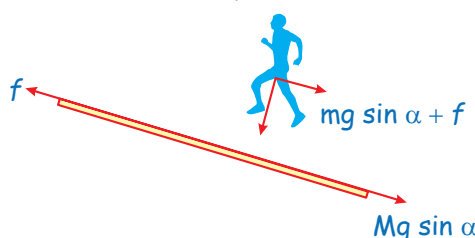


$$\text{From (i) and (ii) } a_{\min} = g \left(\frac{1 - \mu}{1 + \mu} \right)$$

- Q.** A plank is held at an angle α to the horizontal (Fig.) on two fixed supports A and B. The plank can slide against the supports (without friction) because of its weight Mg . With what acceleration and in what direction, a man of mass m should move so that the plank does not move.



Sol. F.B.D. of man and plank are



For plank to be at rest, applying Newton's second law to plank along the incline

$$Mg \sin \alpha = f \quad \dots (i)$$

and applying Newton's second law to man along the incline.

$$mg \sin \alpha + f = ma \quad \dots (ii)$$

$$a = g \sin \alpha \left(1 + \frac{M}{m} \right) \text{ down the incline}$$

Alternate Solution: If the friction force is taken up the incline on man, then application of Newton's second law to man and plank along incline yields.

$$f + Mg \sin \alpha = 0 \quad \dots (i)$$

$$mg \sin \alpha - f = ma \quad \dots (ii)$$

Solving (i) and (ii)

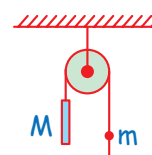
$$a = g \sin \alpha \left(1 + \frac{M}{m} \right) \text{ down the incline}$$

Alternate Solution: Application of Newton's second law to system of man + plank along the incline yields

$$mg \sin \alpha + Mg \sin \alpha = ma$$

$$a = g \sin \alpha \left(1 + \frac{M}{m} \right) \text{ down the incline}$$

- Q.** In the arrangement shown in figure the mass of the rod M exceeds the mass m of the ball. The ball has an opening permitting it to slide along the thread with some friction. The mass of the pulley, mass of the string and the friction in its axle are negligible. At the initial moment the ball was located opposite the lower end of the rod. When set free, both bodies began moving with constant accelerations. Find the friction force between the ball and the thread if t seconds after the beginning of motion the ball got opposite the upper end of the rod. The rod length equals ℓ .



Sol. Step-I: Draw force diagram separately: In fig B, P is a point on the string

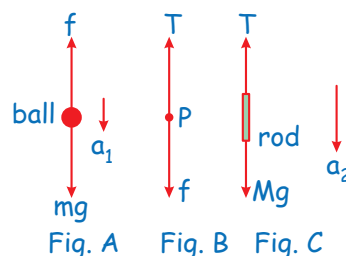
From fig. A,

$$mg - f = ma_1$$

$$\therefore a_1 = \frac{mg - f}{m} \quad \dots (i)$$

From figure B,

$$T = f \quad \dots (ii)$$



From fig. C

$$Mg - T = Ma_2$$

$$Mg - f = Ma_2$$

$$\therefore a_2 = \frac{Mg - f}{M} \quad \dots (iii)$$

Step-II: Apply kinematic relation:

$$s_{\text{rel}} = u_{\text{rel}} t + \frac{1}{2} a_{\text{rel}} t^2 \quad (\text{Shown in fig. D})$$

$$\text{Here } s_{\text{rel}} = \ell, u_{\text{rel}} = 0$$

$$a_{\text{rel}} = a_2 - a_1$$

$$\therefore \ell = \frac{1}{2} (a_2 - a_1) t^2$$

$$\therefore t = \sqrt{\frac{2\ell}{(a_2 - a_1)}}$$

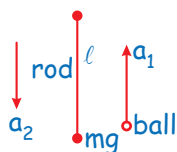
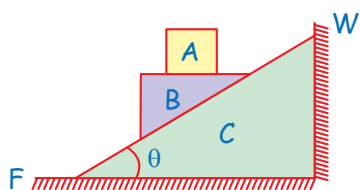


Fig. D

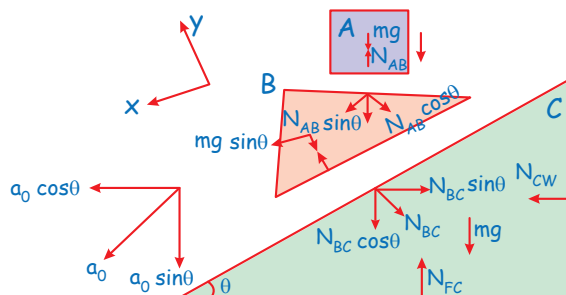
Putting the value of a_1 and a_2 , a_1

$$f = \frac{2\ell Mm}{(M-m)t^2}$$

- Q.** In the figure shown all the surfaces are smooth. The blocks A, B and C have the same mass m . F is floor and W is the wall. Find the magnitude of the contact forces at all the surfaces after the system is released from rest. The angle of inclination of the inclined plane with the horizontal is θ .



Sol.



$$mg - N_{AB} = ma_A$$

[Newton's II law for block A in vertical direction]

$$mg \sin \theta + N_{AB} \sin \theta = ma_B$$

[Newton's II law for block B in x direction]

$$a_A = a_B \sin \theta$$

[Constrained relation for contact surface between block A and B]

Solving above three equations we get

$$N_{AB} = \frac{mg \cos^2 \theta}{1 + \sin^2 \theta}$$

$$mg \cos \theta + N_{AB} \cos \theta - N_{BC} = 0$$

[Equilibrium of block B in y direction]

$$\Rightarrow N_{BC} = mg \cos \theta + \frac{mg \cos^2 \theta \cos \theta}{1 + \sin^2 \theta}$$

$$\Rightarrow N_{BC} = \frac{2mg \cos \theta}{1 + \sin^2 \theta}$$

$$N_{BC} \sin \theta - N_{WC} = 0$$

[Equilibrium of block C in horizontal direction]

$$N_{WC} = \frac{2mg \sin \theta \cos \theta}{1 + \sin^2 \theta}$$

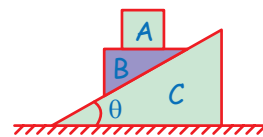
$$N_{BC} \cos \theta + mg - N_{FC} = 0$$

[Equilibrium of block C in vertical direction]

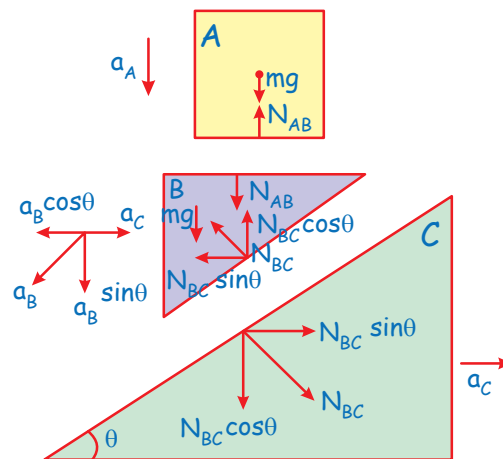
$$\Rightarrow N_{FC} = \frac{2mg \cos^2 \theta}{1 + \sin^2 \theta} + mg$$

$$\Rightarrow N_{FC} = \frac{mg(2 + \cos^2 \theta)}{1 + \sin^2 \theta}$$

- Q.** In the figure shown all blocks are of equal mass ' m '. All surfaces are smooth. Find the acceleration of all the blocks.



Sol.



a_A and a_C are acceleration of block A and C w.r.t ground and a_B is the acceleration of block B w.r.t. block C

$$N_{BC} \sin \theta = ma_C$$

[Newton's II law for block C in horizontal direction in ground frame]

$$N_{AB} + mg - N_{BC} \cos \theta = ma_B \sin \theta$$

[Newton's II law for block B in vertical direction in ground frame]

$$N_{BC} \sin \theta = m(a_B \cos \theta - a_c)$$

[Newton's II law for block B in horizontal direction in ground frame]

$$mg - N_{AB} = ma_A$$

[Newton's II law for block A in vertical direction in ground frame]

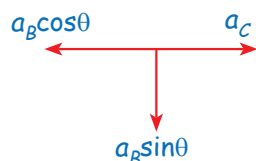
$$a_A = a_B \sin \theta$$

[constrained relation for block A and B in vertical direction]

Now we have 5 equations and 5 unknowns.

$$\Rightarrow a_A = \frac{4g \sin^2 \theta}{1 + 3 \sin^2 \theta} \Rightarrow a_B = \frac{4g \sin \theta}{1 + 3 \sin^2 \theta}$$

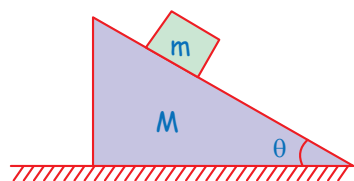
$$\Rightarrow a_c = \frac{2g \sin \theta \cos \theta}{1 + 3 \sin^2 \theta}$$



$\Rightarrow$ Acceleration of B w.r.t ground

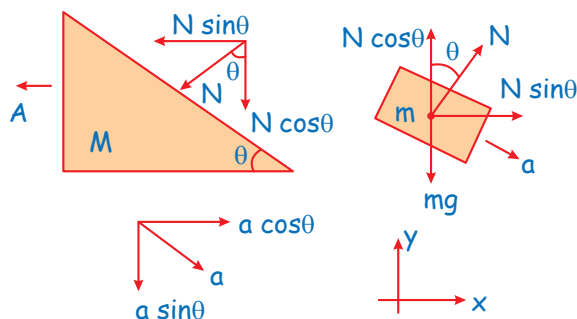
$$= \sqrt{(a_B \cos \theta - a_c)^2 + (a_B \sin \theta)^2} = \frac{2g \sin \theta}{\sqrt{1 + 3 \sin^2 \theta}}$$

Q. A block of mass m is placed on the inclined surface of a wedge as shown. Calculate the acceleration of the wedge and the block when the block is released. Assume all surfaces are frictionless.



Sol. Let the acceleration of wedge be A and that of block be a (w.r.t. wedge). Then acceleration of m w.r.t. ground is

$$(a \cos \theta - A) \hat{i} - a \sin \theta \hat{j}$$



$$\text{For } M: N \sin \theta = MA \quad \dots (i)$$

$$\text{For } m: mg - N \cos \theta = ma \sin \theta \quad \dots (ii)$$

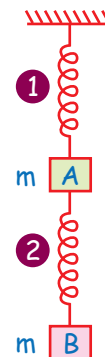
$$N \sin \theta = m(a \cos \theta - A) \quad \dots (iii)$$

Solved (i), (ii) and (iii) to get

$$A = \frac{mg \sin \theta \cos \theta}{M + m \sin^2 \theta} \text{ and } a = \frac{(M + m)g \sin \theta}{M + m \sin^2 \theta}$$

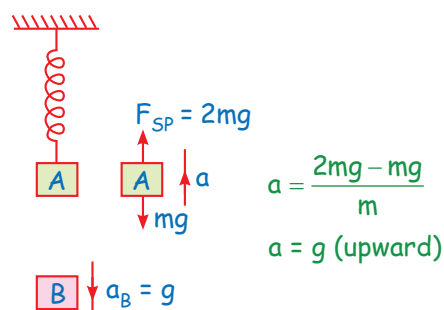
Now acceleration of block can be found by putting the values of A and a in (i).

Q. Find acc of each block just after spring cutting if



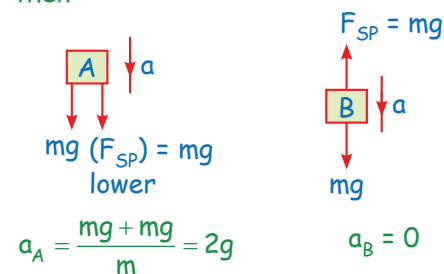
(a) Lower spring is cut

$(F_{SP})_1 = 2mg \rightarrow$ will not change suddenly
hence

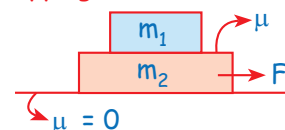


$$a = \frac{2mg - mg}{m} = g \text{ (upward)}$$

(b) If upper spring cut at $t = 0$ then



Q. Find $F_{\max}$ so that both block move together without slipping between them



Sol. $(a_{\text{upper}})_{\text{max}} \rightarrow$ given by friction only

$$(a_{\text{upper}})_{\text{max}} = \frac{f_{\text{max}}}{m_1} = \frac{\mu m_1 g}{m_1}$$

$$(a_{\text{upper}})_{\text{max}} = \mu g$$

$(a_{\text{lower}})_{\text{max}}$ so that both will move with same acc = μg

$$\Rightarrow F_{\text{max}} = (m_1 + m_2)a = (m_1 + m_2)\mu g$$

Q. F_{max} so that both move with same acc.

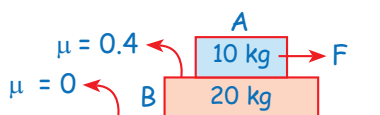


Sol. $(a_A)_{\text{max}} = \frac{f_{\text{max}}}{m_A} = \frac{0.4 \times 100}{10} = 4$

$$F_{\text{max}} = (10 + 20) \times 4 = 120 \text{ N}$$

If $F \leq 120$ both move with same acc.

Q. Find F_{max} so that both move with same acc.



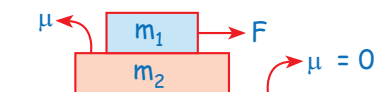
Sol. $(a_B)_{\text{max}} = \frac{f_{\text{max}}}{m_B} = \frac{0.4 \times 100}{20} = 2$

$$F_{\text{max}} = (10 + 20) \times 2 = 60 \text{ N}$$

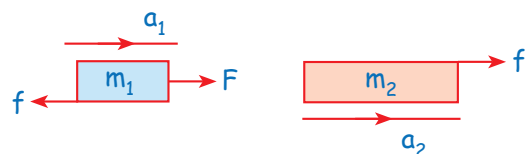
If $F > 60 \text{ N}$ Both move with difference acc

If $F \leq 60$ Both move with same acc.

Q. Find F_{max} so that both block move with same acc.



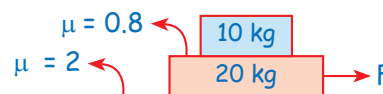
Sol.



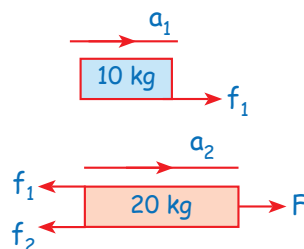
$$(a_2)_{\text{max}} = \frac{f_{\text{max}}}{m_2} = \frac{\mu m_1 g}{m_2}$$

$$\Rightarrow F_{\text{max}} = (m_1 + m_2) \frac{\mu m_1 g}{m_2}$$

Q. Find F_{max} so that both move with same acc.

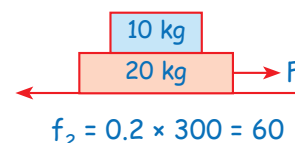


Sol.



$$(a_1)_{\text{max}} = \frac{(f_1)_{\text{max}}}{m_1} = \frac{0.8 \times 100}{10} = 8$$

$$(a_{\text{common}})_{\text{max}} = 8$$



$$f_2 = 0.2 \times 300 = 60$$

$$F - f_2 = (10 + 20) a_{\text{common}}$$

$$F - 60 = 30 \times 8$$

$$F = 300 = F_{\text{max}}$$

$F > 300$ both block will move with diff acc.

$60 < F \leq 300$ both block will move with same acc.

$$F \leq 60 \quad a_1 = a_2 = 0$$

Q. In the above question find the acceleration of both the blocks when

(i) $F = 18 \text{ N}$

(ii) $F = 36 \text{ N}$

Sol. (i) Since $F < 30$ both the blocks will move together

F.B.D

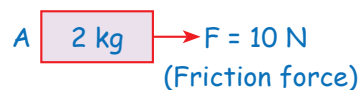


$$a = \frac{18}{6} = 3 \text{ m/s}^2$$

(ii) $F = 36 \text{ N}$

When $F > 30$ both the blocks will move separately so we treat each block independently

F.B.D of 2 kg block



$$a_A = 5 \text{ m/s}^2$$

F.B.D of 4 kg block

$$F = 10 \text{ N} \leftarrow \boxed{\text{4 kg}} \rightarrow F = 36 \text{ N}$$

$$a_B = \frac{36 - 10}{4} = \frac{26}{4} \text{ m/s}^2$$

Q. Friction coefficient between the blocks is 0.5 and ground is smooth. If force of 120 N is applied on the upper block as shown in figure find acceleration of the blocks.



Sol. $f_{\max} = 0.5 \times 10g = 50 \text{ N}$

Let's assume they have same acceleration. The value of common acceleration is $\frac{120}{30} = 4 \text{ m/s}^2$.

Making free-body diagram



Newton's second law given

$$f = 20 \times 4 = 80 \text{ N}$$

Solving we get, $a = 4 \text{ m/s}^2$ and $f = 80 \text{ N}$ while limiting value of friction is 50 N. hence our assumption is wrong.

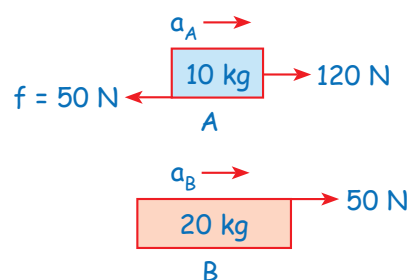
Let's say they are not moving together. Then $f_k = 50 \text{ N}$.

By Newton's second law.

$$120 - 50 = 10 a_A$$

$$50 = 20 a_B$$

$$a_A = 7 \text{ m/s}^2, a_B = 2.5 \text{ m/s}^2$$



This gives us the correct direction of friction.